Lecture 1:

Introduction

Computer Graphics and Imaging
UC Berkeley CS184/284A

Welcome to CS184 / 284A!



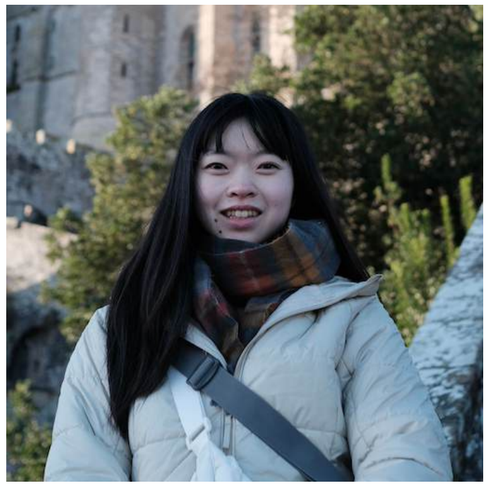
Prof. James F. O'Brien

- Ph.D. GaTech 2000 on fracture propagation with application to destruction visual effects.
- Research: Computer graphics, computer animation, simulations of physical systems, human perception, rendering, image synthesis, machine learning, virtual reality, and media forensics.
- Industry: Kronos, Pixelux, Dreamworks, Avametric, Juice Technologies
- Enjoys: Photography, traveling, scuba, camping, beat saber, woodworking, and glassblowing.

Welcome to CS184 / 284A!



Sriram Srivatsan



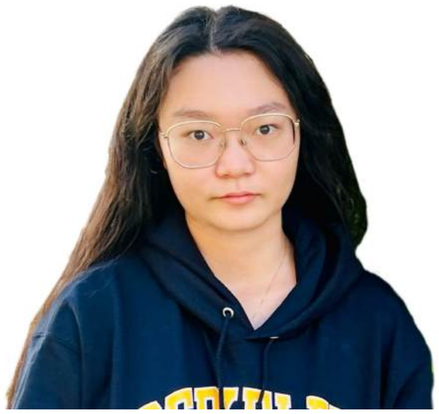
Jessica Lee



Swetha Rajkumar



Atsunobu Kotani



Karina Jin

Arnav Dalal

Your Names: An Important Request

We want to get to know you

It starts with your names

We want to remember, but there are many of us

Please help us with this rule:

- Every time you participate in class, section, office hours, discussion post, etc. please remind us your name.

Example: “Hi, this is Susan. My question is about…”

Thank you very much!

CS184/284A: Computer Graphics & Imaging

Why Study Computer Graphics?

Course Overview

Logistics

Prerequisites

Math

- Vectors, matrices, basic linear algebra
- Helpful: exposure to statistics, signal processing, Fourier transform

Programming

- Data structures (CS61B)
- Fluent with C and C++
- Fluent with development environment, debugging, etc.

Use of AI

The use of AI tools is permitted and encouraged!

- Use them to *help you learn*
- Ask for advice and explanations
- Use AI to save time, not to avoid learning

Philosophy: Education is like going to the gym

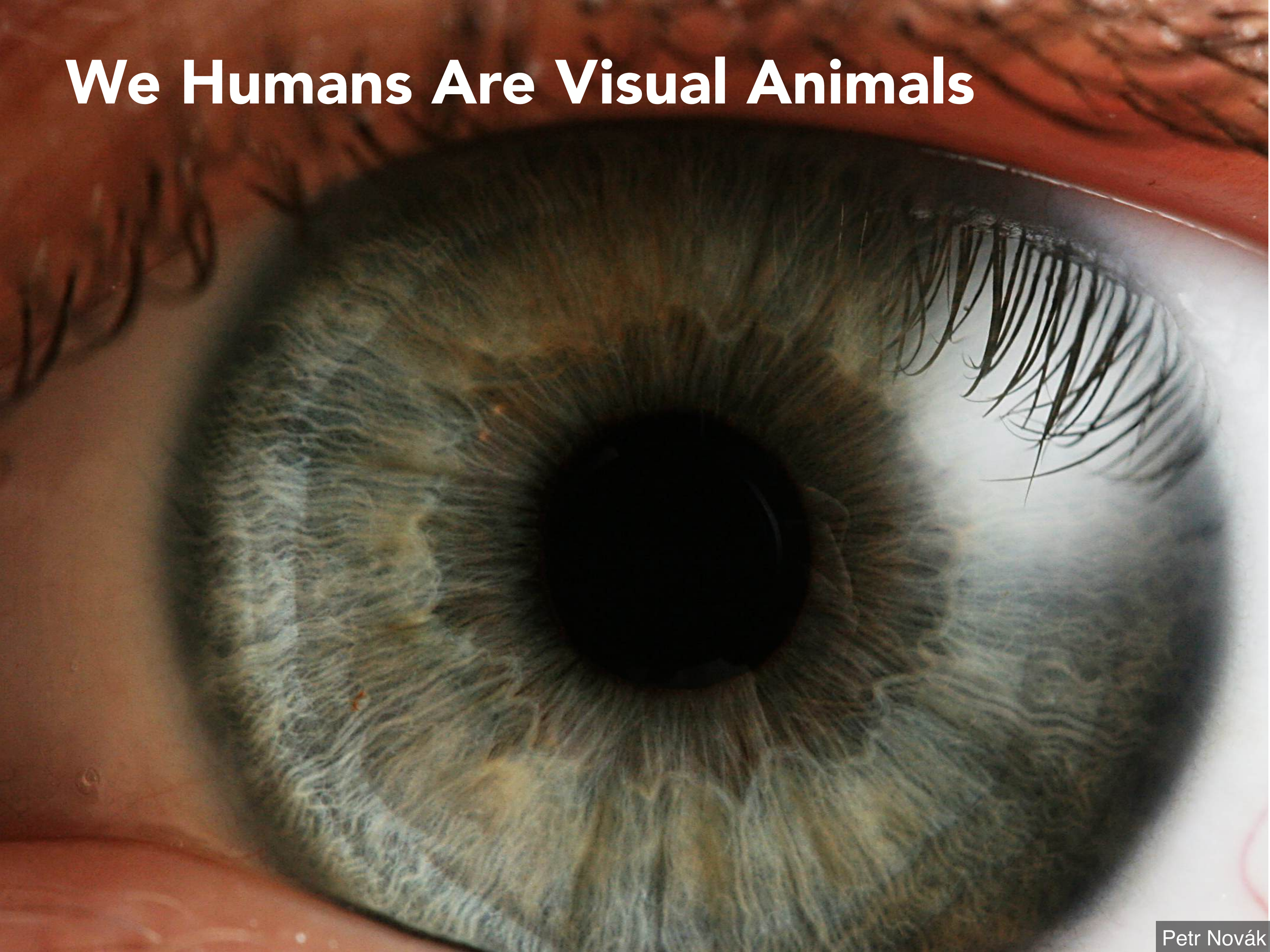
- You pay time and money for the opportunity
- No results without honest effort
- Ultimately it is about what you do for yourself

What is Computer Graphics?

com•put•er graph•ics /kəmˈpyo͞odər ˈgrafiks/ n.
The use of computers to synthesize and manipulate visual information.

Why Visual Information?

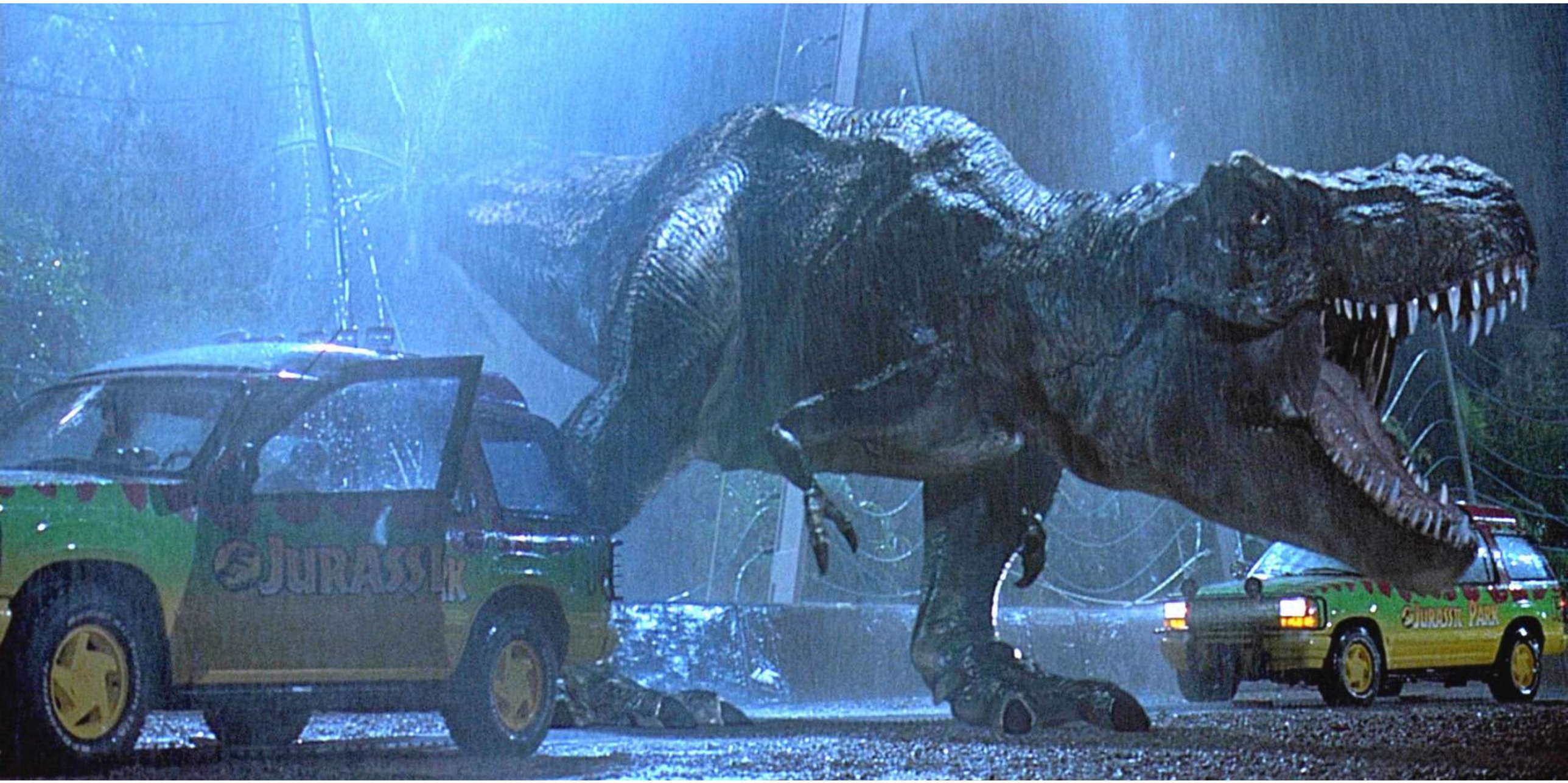
We Humans Are Visual Animals



Petr Novák

Why Study Computer Graphics and Imaging?

Movies



Jurassic Park (1993)



Moments That Changed The Movies: Jurassic Park
https://www.youtube.com/watch?v=KWsbcBvYqN8

Movies



Toy Story (1995)

Movies



The Matrix (1999)

Movies









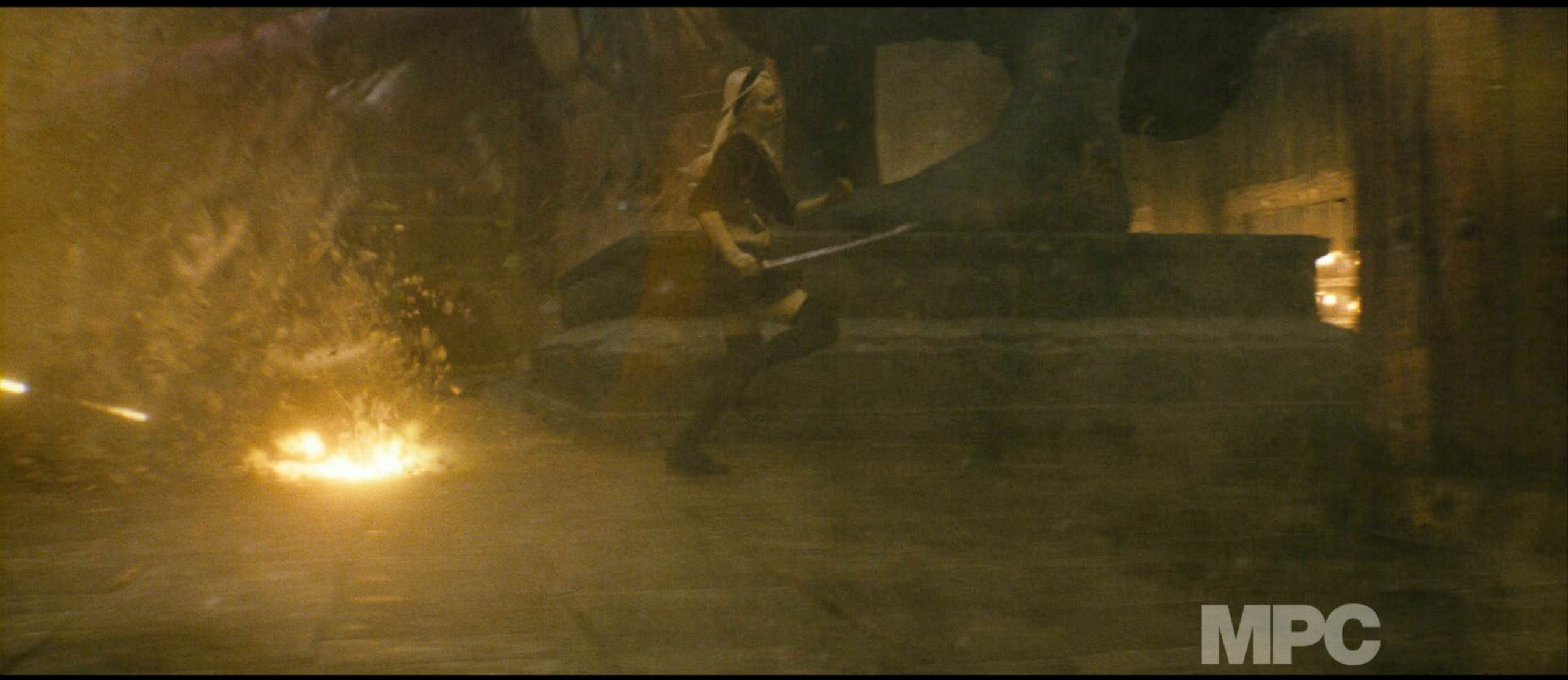
The Matrix (1999)

The Campanile



Debevec, Taylor and Malik SIGGRAPH 1996

https://www.pauldebevec.com/Campanile/



Sucker Punch (2009)

Motion Capture



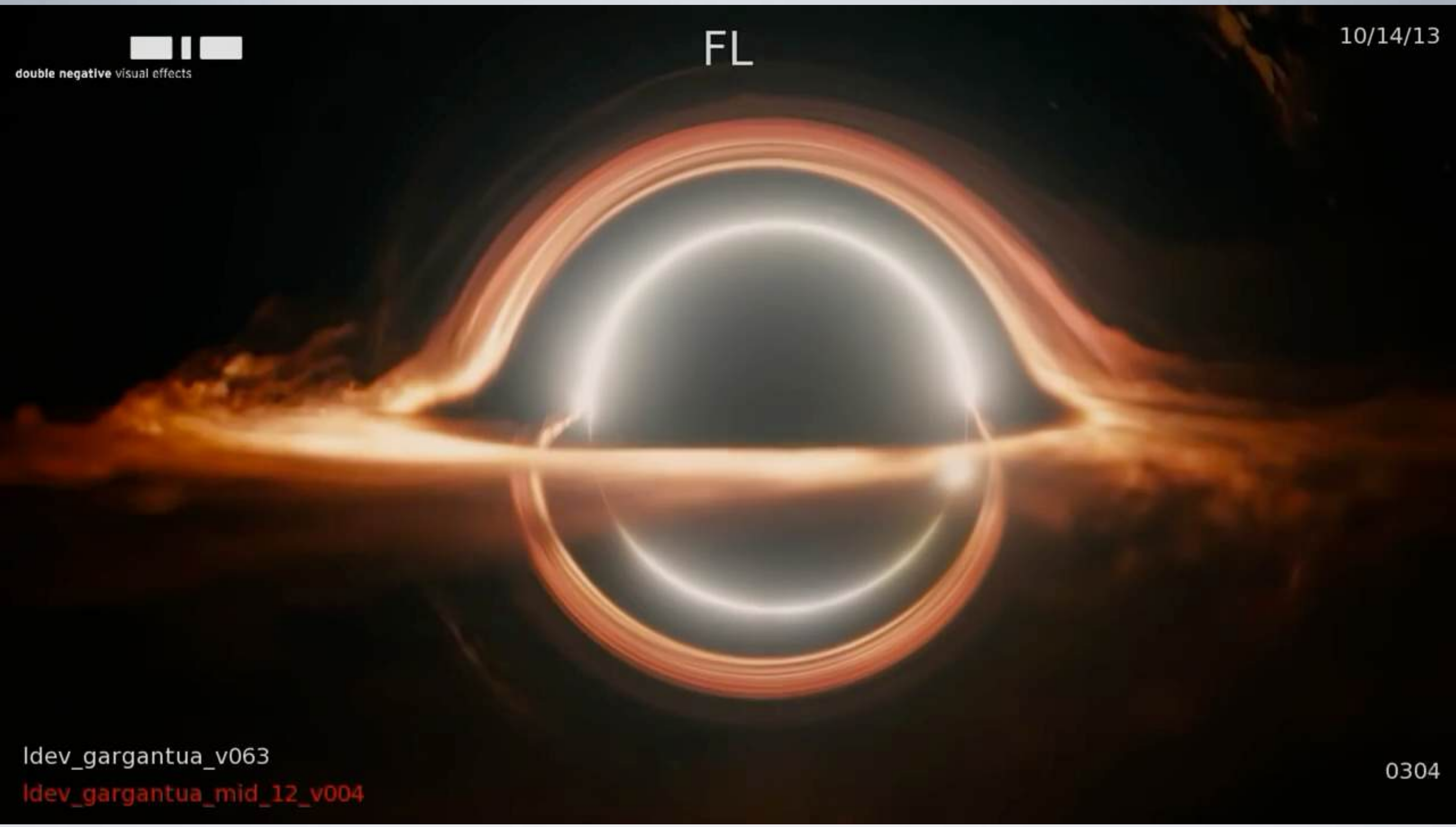
Andy Serkis in The Two Towers

CS184/284A

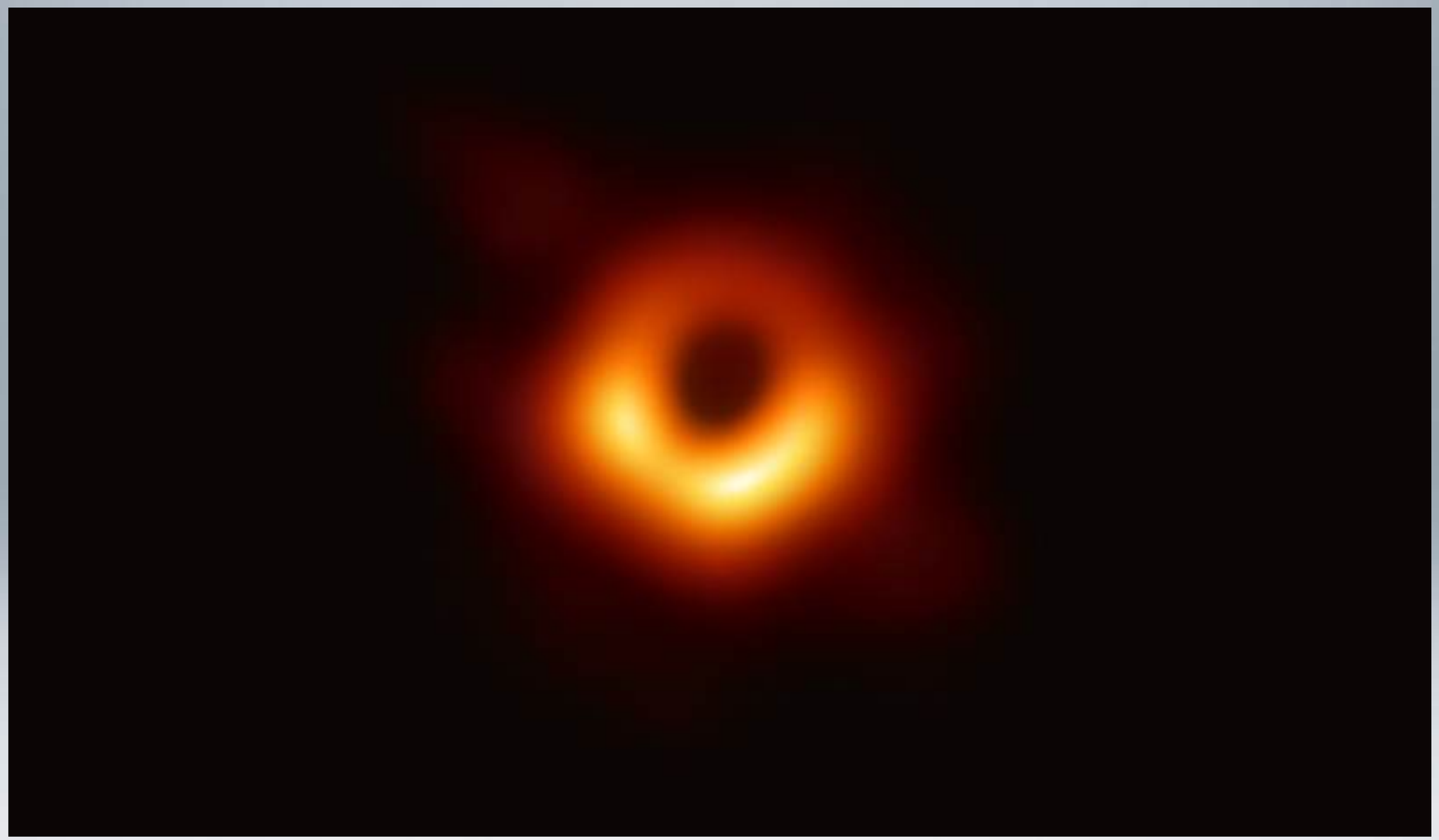
Ren Ng

Avatar (2009)

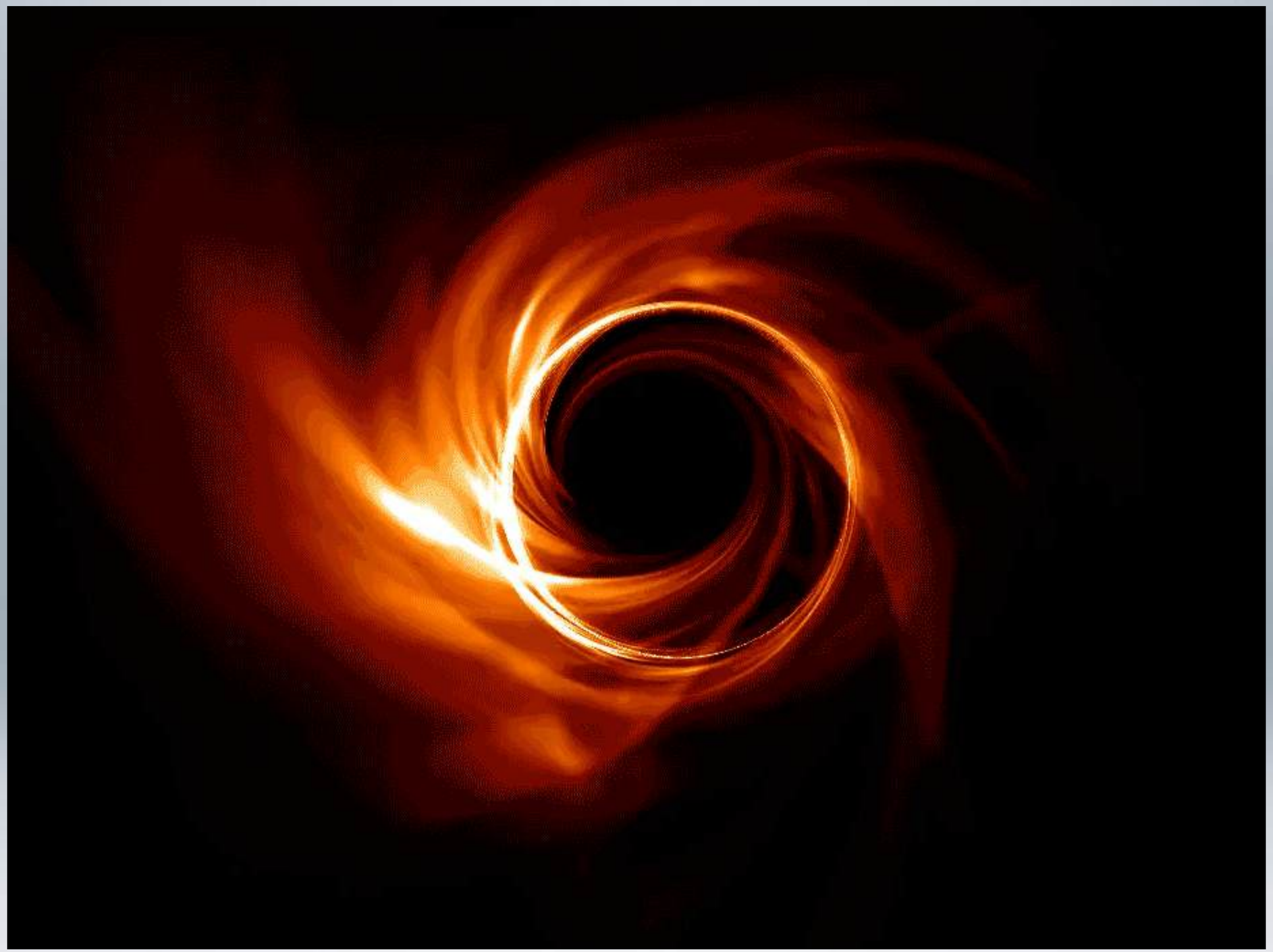




Interstellar (2014)

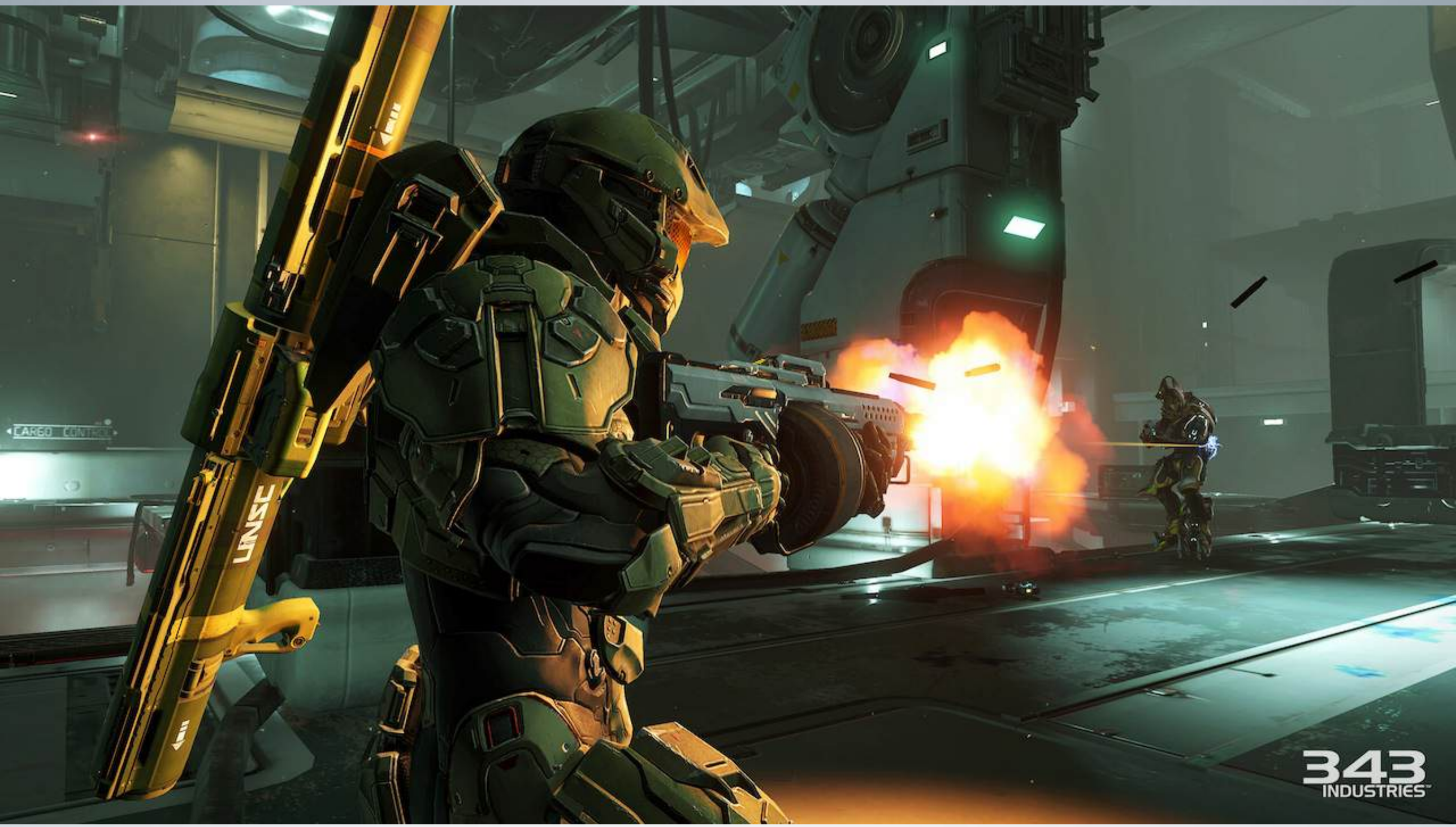


Event Horizon Telescope collaboration et al.



Event Horizon Telescope collaboration et al.

Games



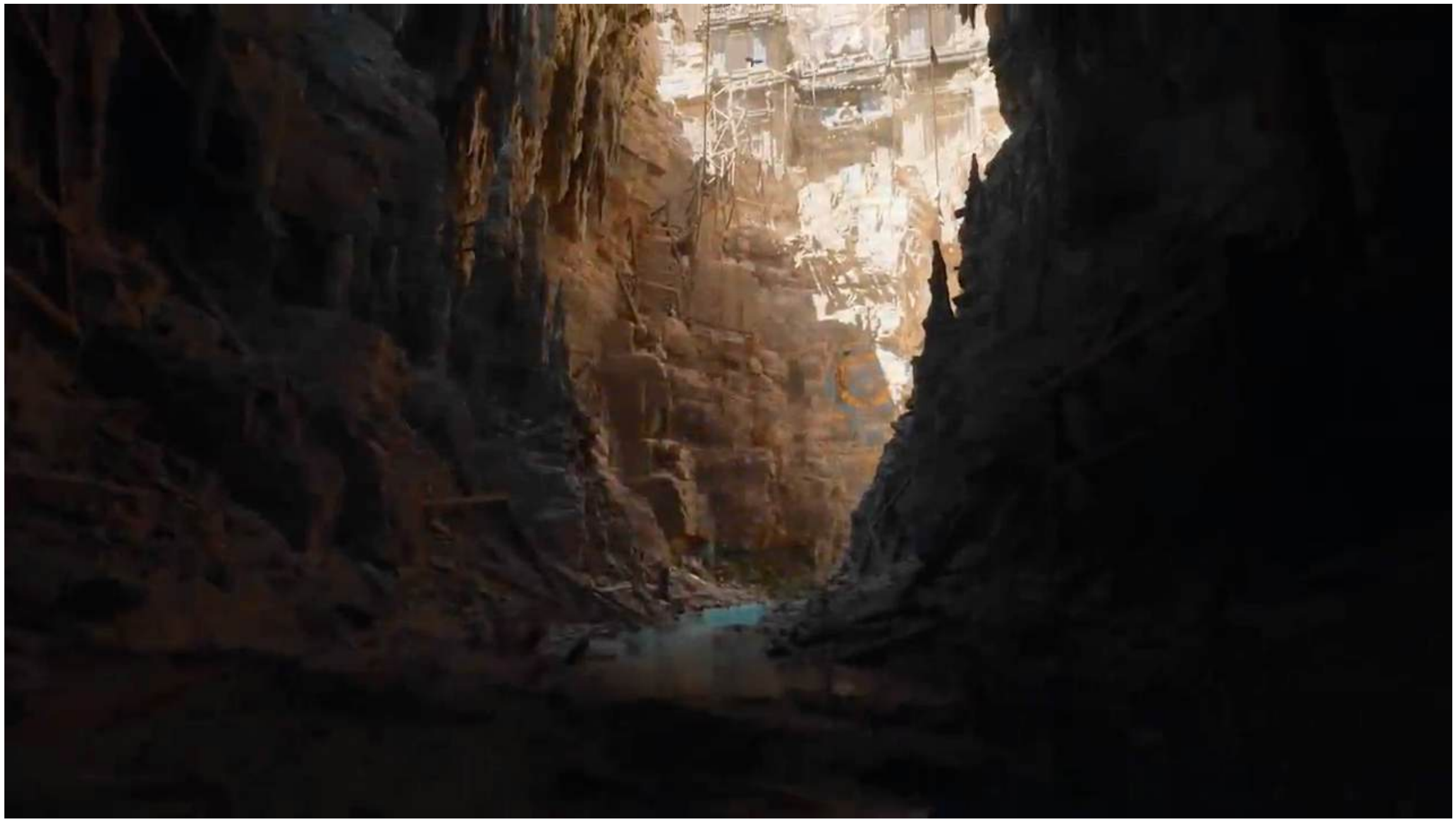
Halo 5 (2015)

Games



Super Mario World

Games



Unreal Engine 5 Demo Realtime in PS5 (2020)

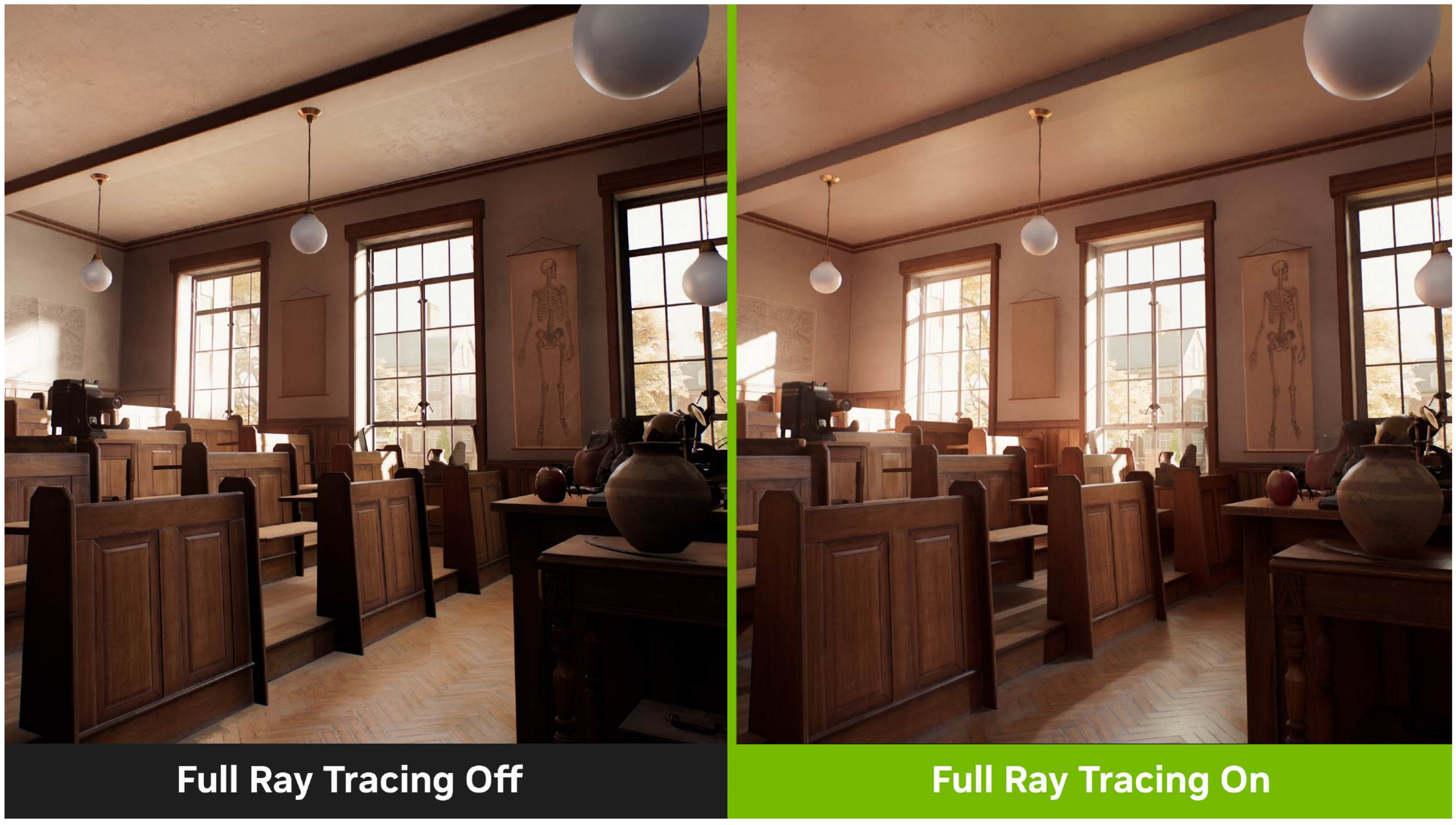


Battlefield 5 (2018)



Battlefield 5 (2018)

Games



Indiana Jones and the Great Circle (2024)

Realtime ray tracing (path tracing) for global illumination

Games



Mirror's Edge (2008)



Ōkami (2006)

Training Simulations



From America's Army



From CAE Inc.

Visual Simulation



Driving simulator
Toyota Higashifuji Technical Center



da Vinci surgical robot
Intuitive Surgical

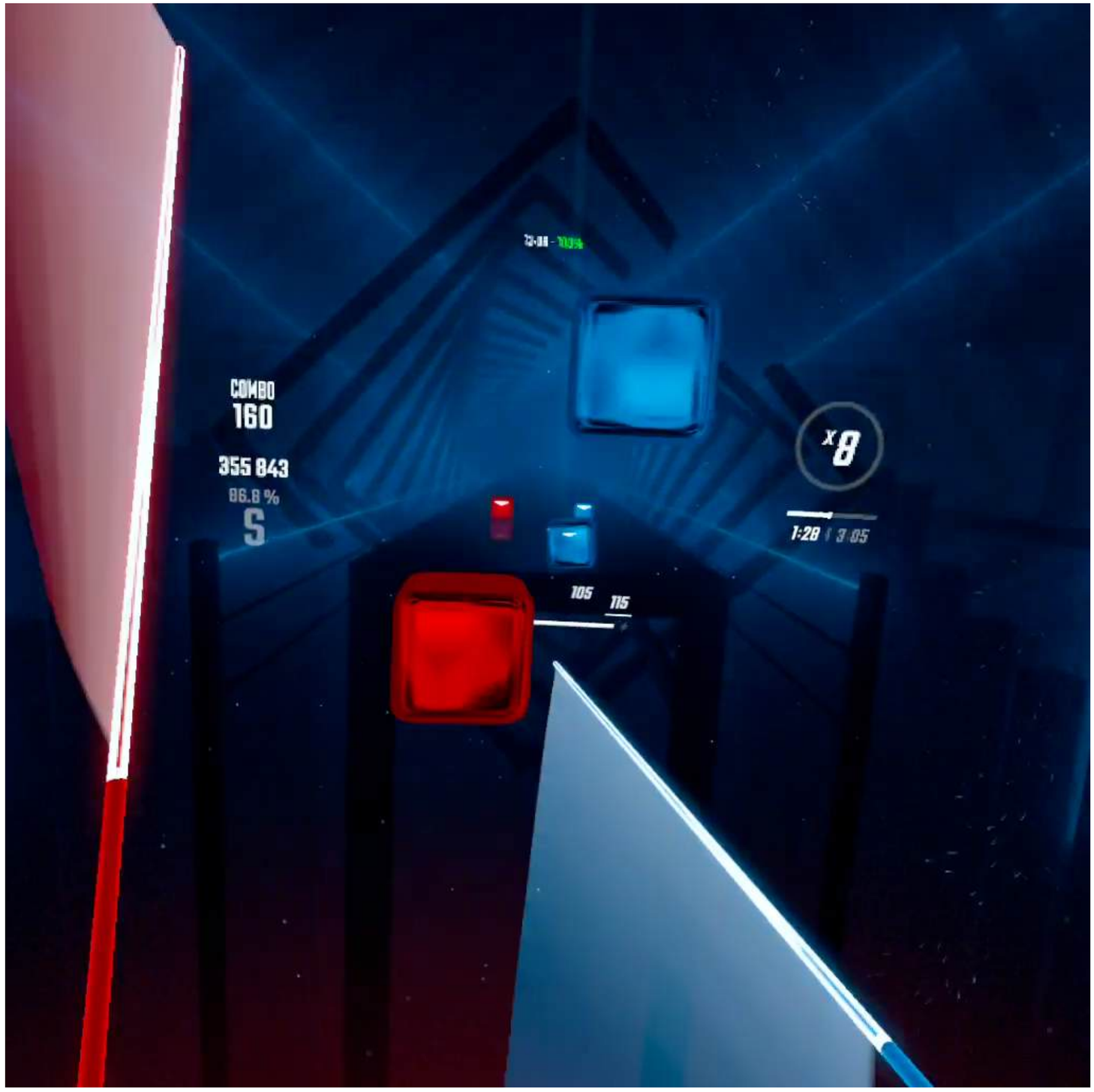
Flight simulator, driving simulator, surgical simulator, …

Virtual Reality



HTC Vive headset and controllers

Virtual Reality



Augmented Reality

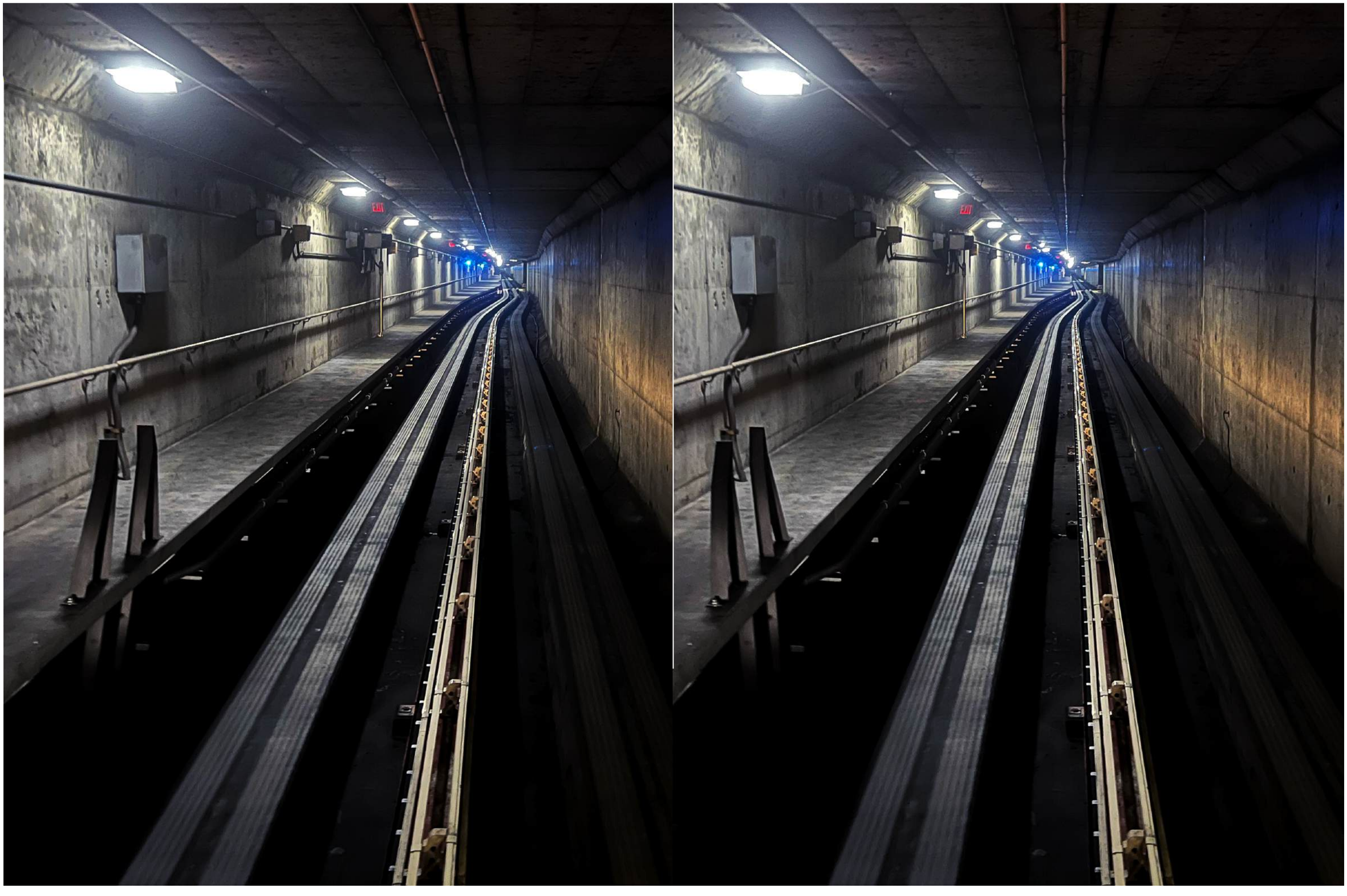


Microsoft Hololens augmented reality headset concept

Augmented Reality



Stereo Vison



Product Design and Visualization



Ikea - 75% of catalog is rendered imagery

Product Design and Visualization



Photograph



Simulation

Avametric 2016

Product Design and Visualization



Tesla Model X concept (2012)

Product Design and Visualization



Mill Blackbird

Architectural Design



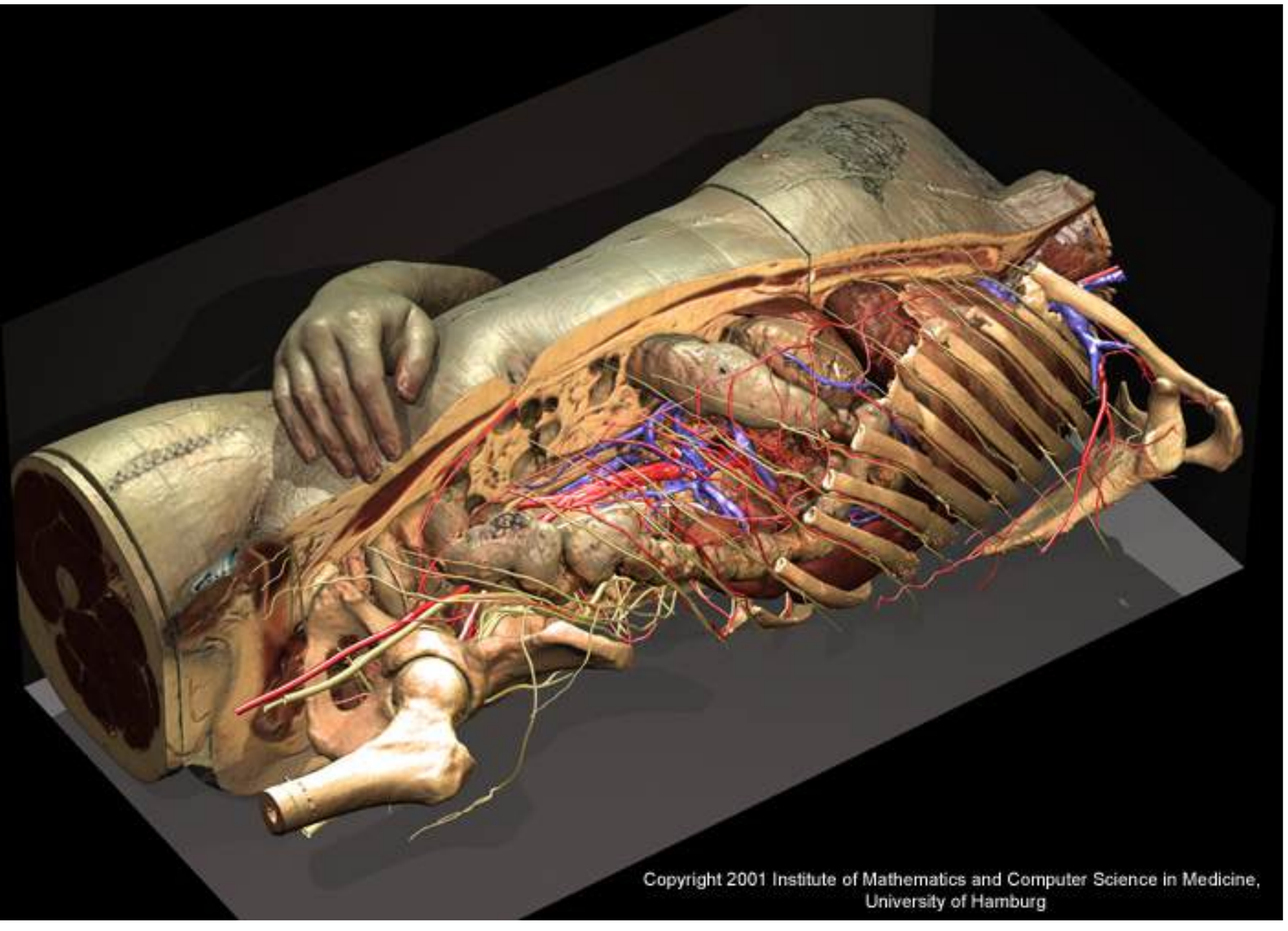
Bilbao Guggenheim, Frank Gehry

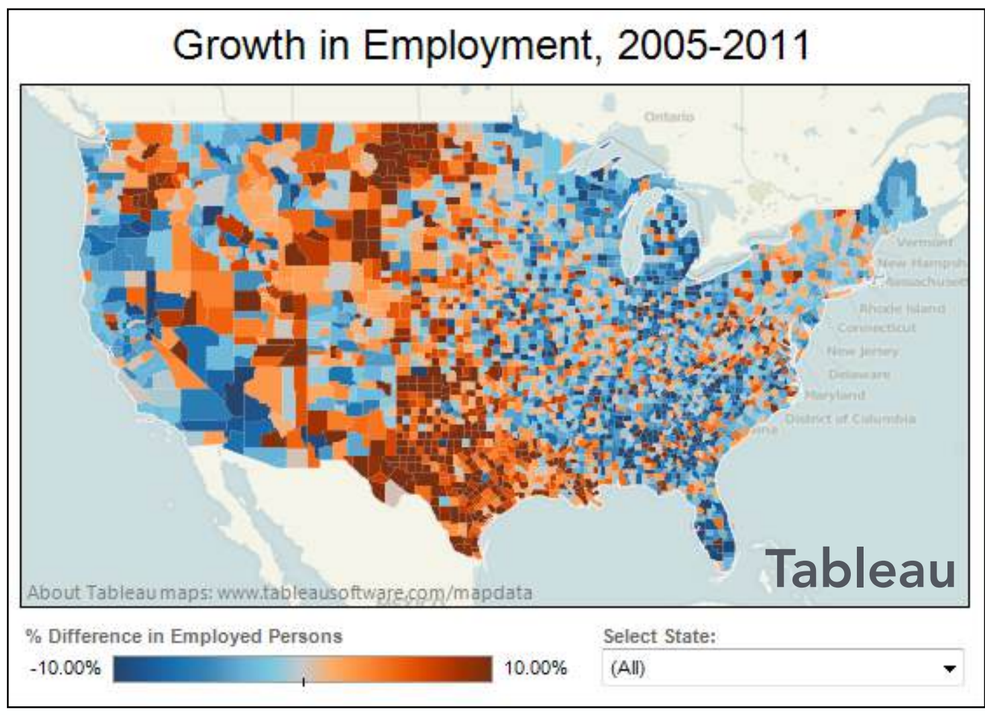
Architectural Design



Heydar Aliyev Center, Zaha Hadid Architects

Visualization







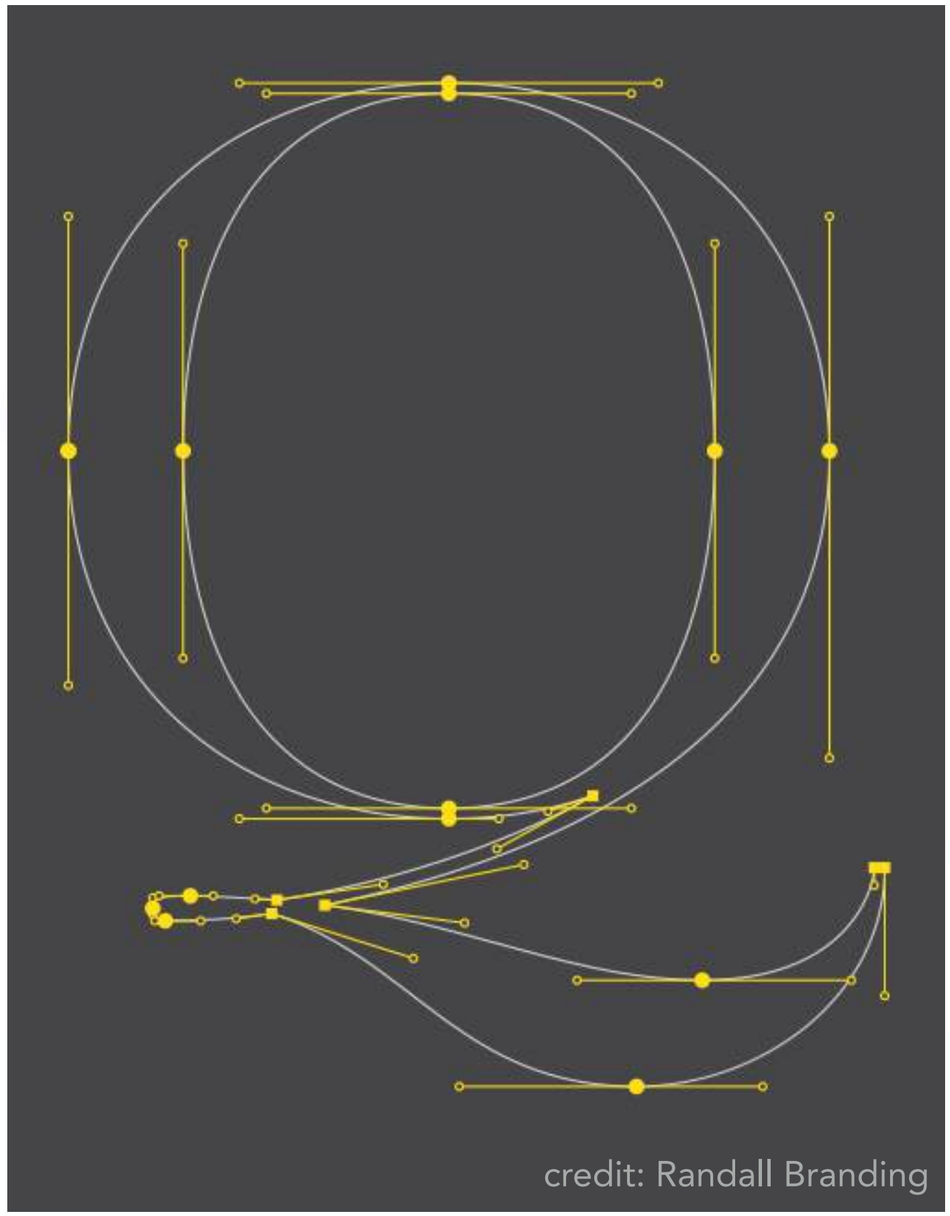


Science, engineering, medicine, journalism, …

Typography

The Quick Brown Fox Jumps Over The Lazy Dog

ABCDEFGHIJKLMNOPQRSTUVWXYZ
abcdefghijklmnopqrstuvwxyz 0123456789



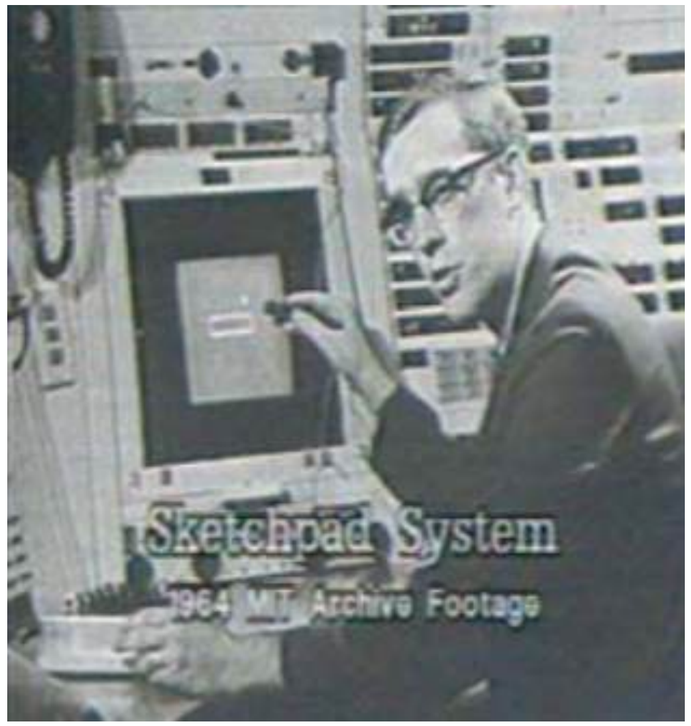
credit: Randall Branding

Baskerville

Graphical User Interfaces

Desktop metaphor

- Input: Keyboard, mouse
- Ouput: Cathode-ray tube



Ivan Sutherland, Sketchpad
Light pen, vector display



Doug Engelbart
Mouse

Graphical User Interfaces





2D drawing and animation are ubiquitous in computing.
Typography, icons, images, transitions, transparency, ...

Graphical User Interfaces



2D drawing and animation are ubiquitous in computing.

Typography, icons, images, transitions, transparency, ...

Graphical User Interfaces



https://www.youtube.com/watch?v=YndL315tQq8

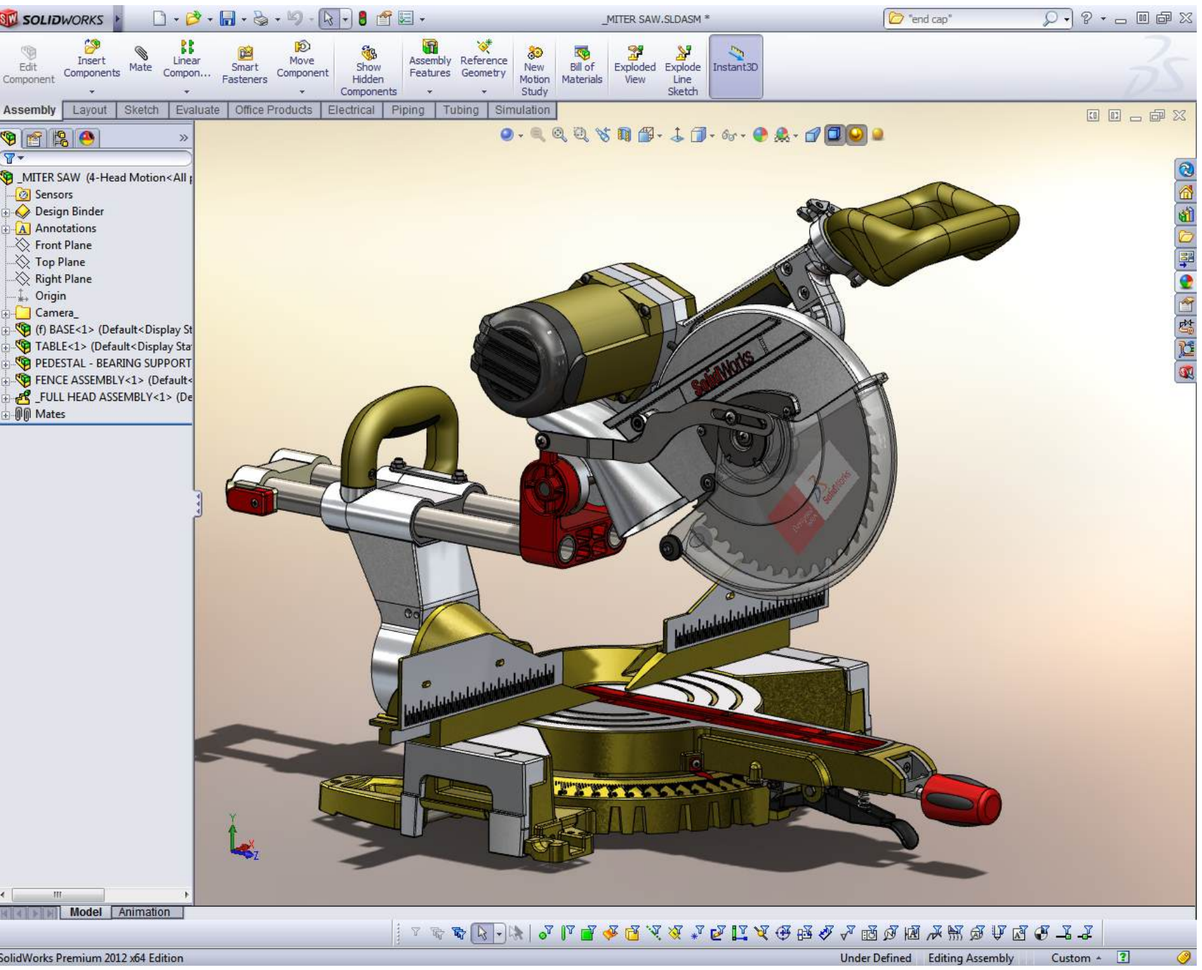
Digital Illustration



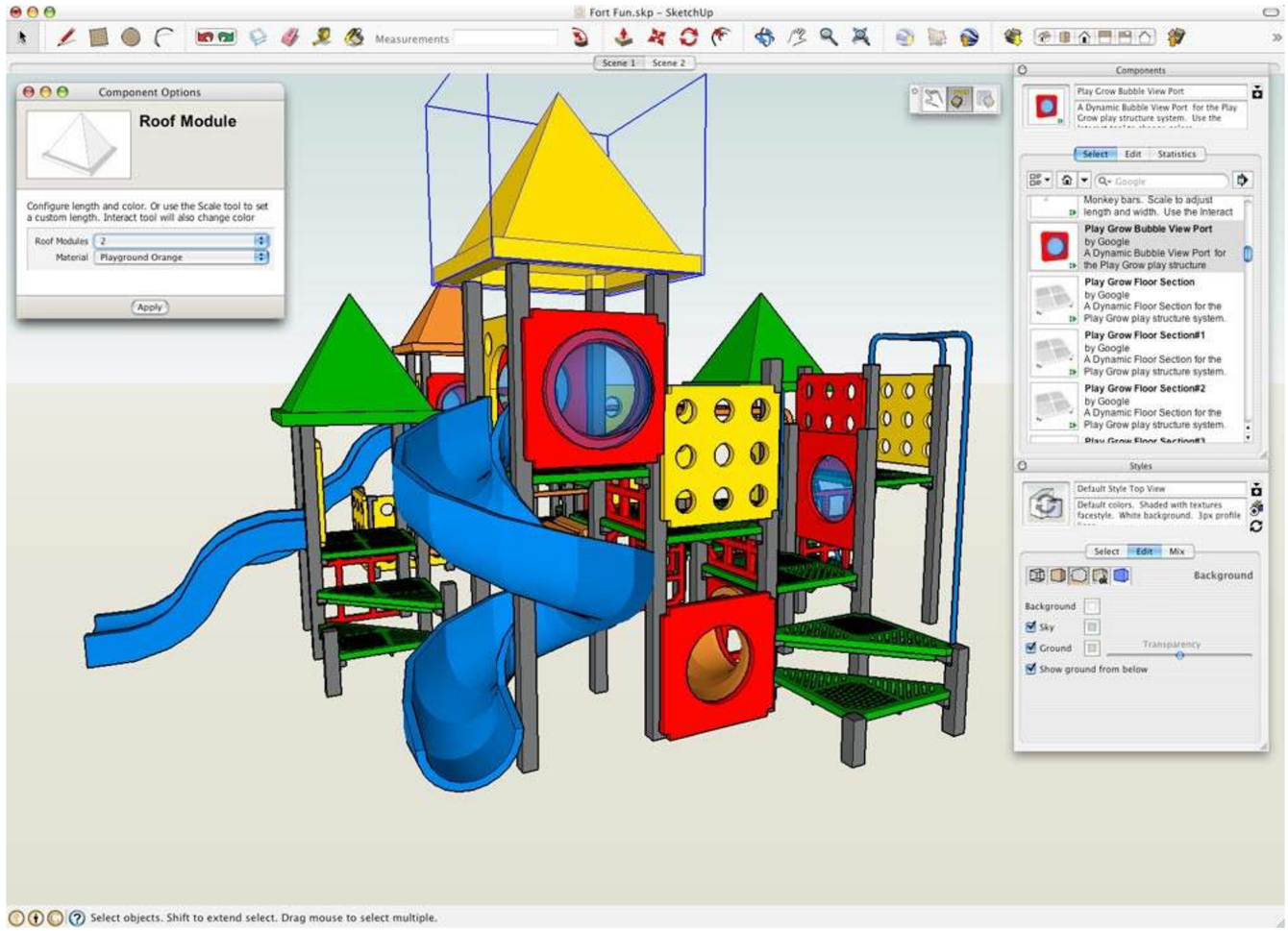
Meike Hakkart

http://maquenda.deviantart.com/art/Lion-done-in-illustrator-327715059

Computer-Aided Design



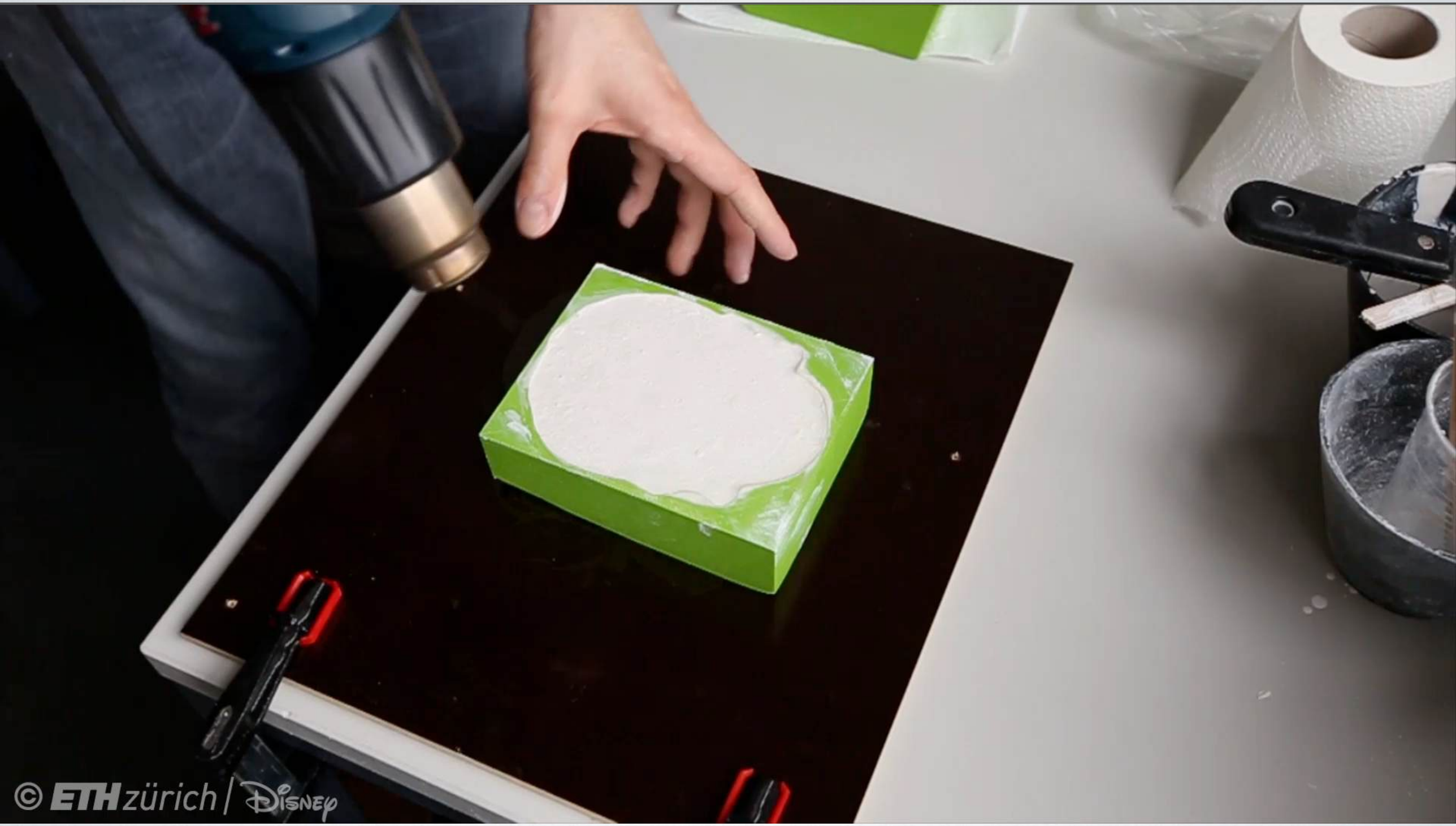
SolidWorks



SketchUp

For mechanical, architectural, electronic, optical, …

Fabrication



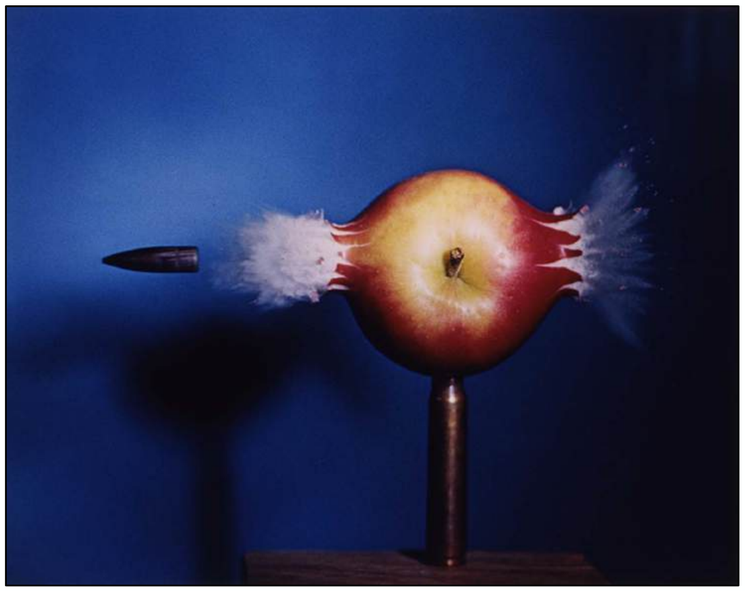
Computational Thermoforming
Christian Schuller, Daniele Panozzo, Anselm Grundhofer, Henning Zimmer, Evgeni Sorkine, Olga Sorkine-Hornung

Photography

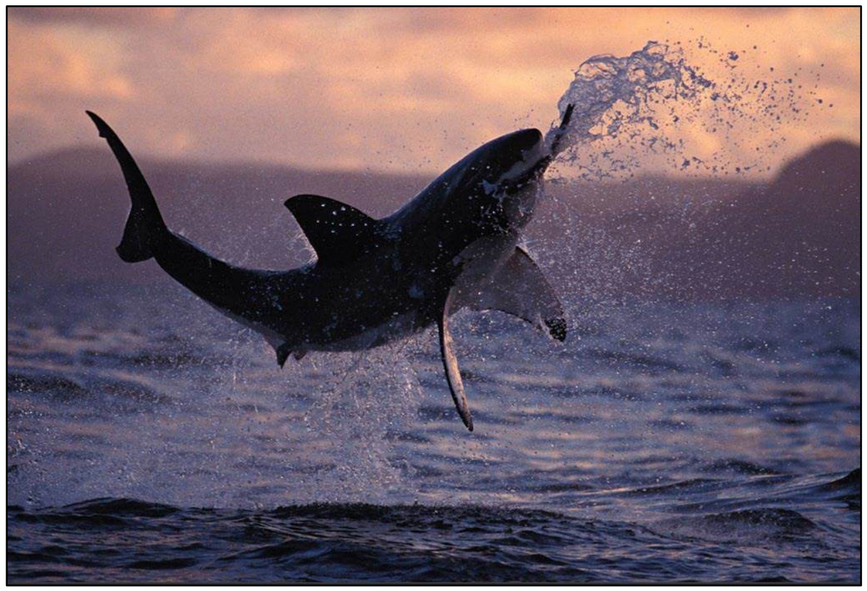










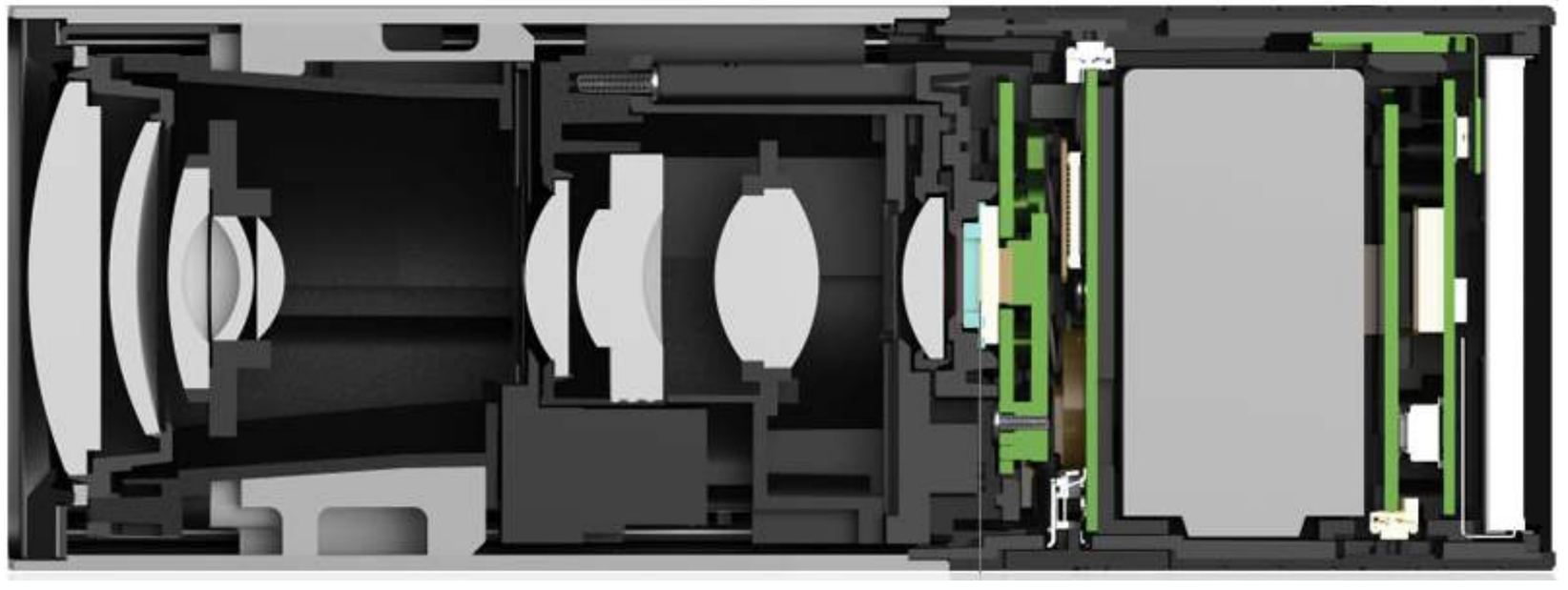


NASA | Walter Iooss | Steve McCurry
Harold Edgerton | NASA | National Geographic

Digital and Computational Cameras







Panaromic stitching, HDR photos, light field cameras, …

Ubiquitous Imaging



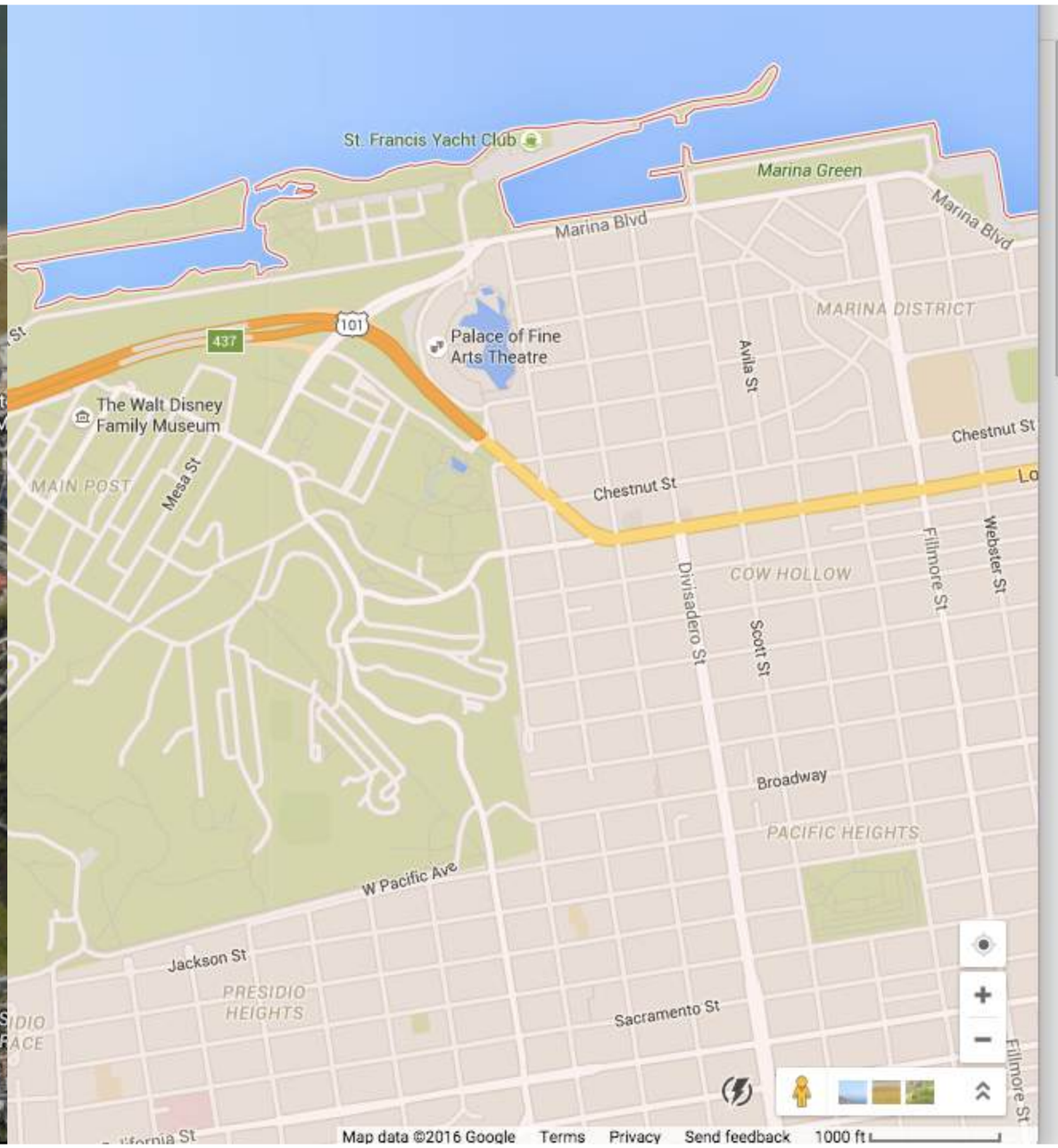




Cameras everywhere

Imaging in Mapping





Maps, satellite imagery, street-level imaging,...

Imaging in Mapping



Maps, satellite imagery, street-level imaging,...

Imaging for Computer Vision



ImageNet: 15M images, 22K categories

http://image-net.org

Imaging for Robotics



Google's "Arm Farm"

Foundations of Graphics and Imaging

These applications require sophisticated theory and systems

Science and Mathematics

- Physics of light, color, optics, …
- Math of curves, surfaces, geometry, perspective, …

Technology and Systems

- Input devices, GPUs, displays, …
- Cameras, lenses, sensors, …

Art and Psychology

- Perception: color, stereo, motion, image quality, …
- Art and design: composition, form, lighting, …

Course Goals

Overview of core ideas in graphics and imaging

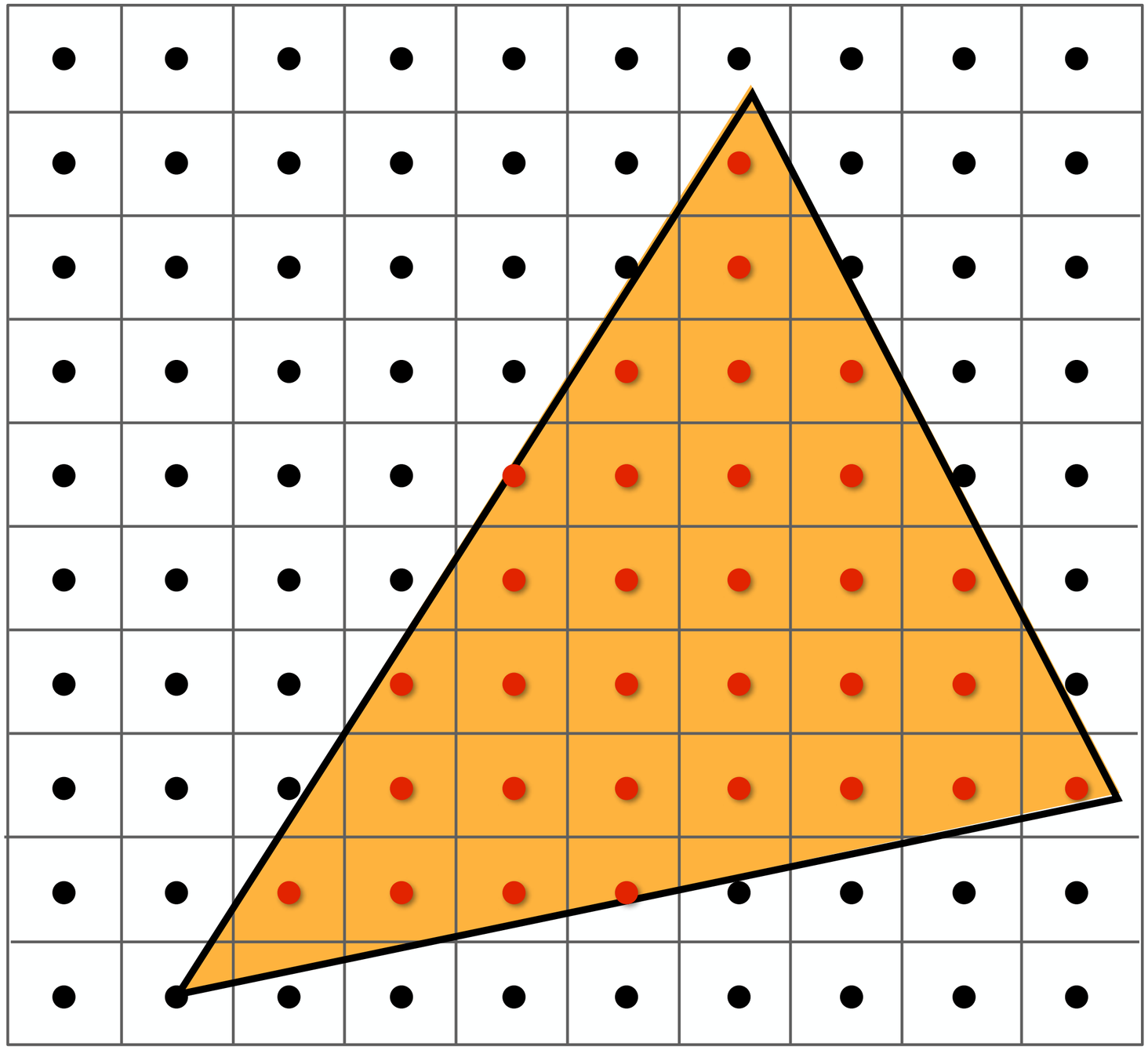
- Modeling the world, image synthesis
- 3D graphics: geometry, rendering, animation
- Image capture, manipulation and display

Acquire core concepts and skills

- Representations (geometry, images, transforms, …)
- Algorithms (sampling, subdivision, ray-tracing, …)
- Technology (GPUs, displays, cameras, …)

Course Topics

Drawing Digital Images (Rasterization)



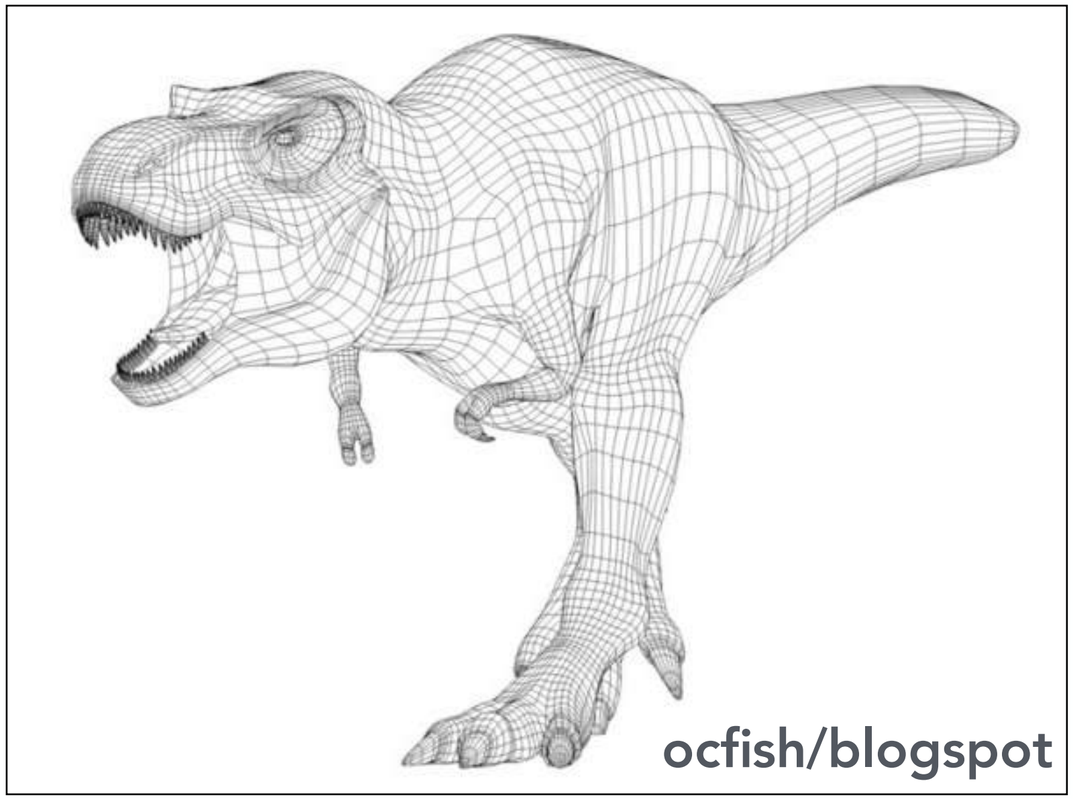
Filtering and Sampling



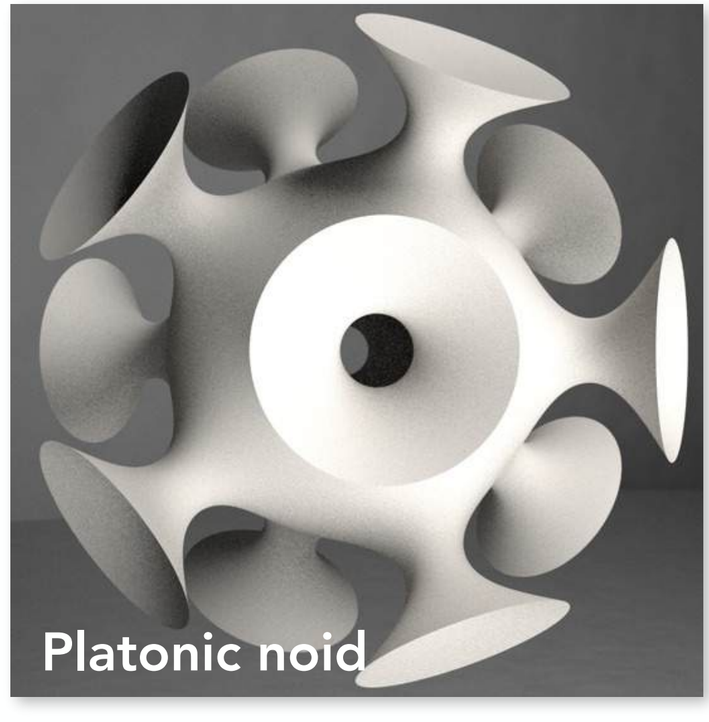
No Jaggies

Modeling Geometry



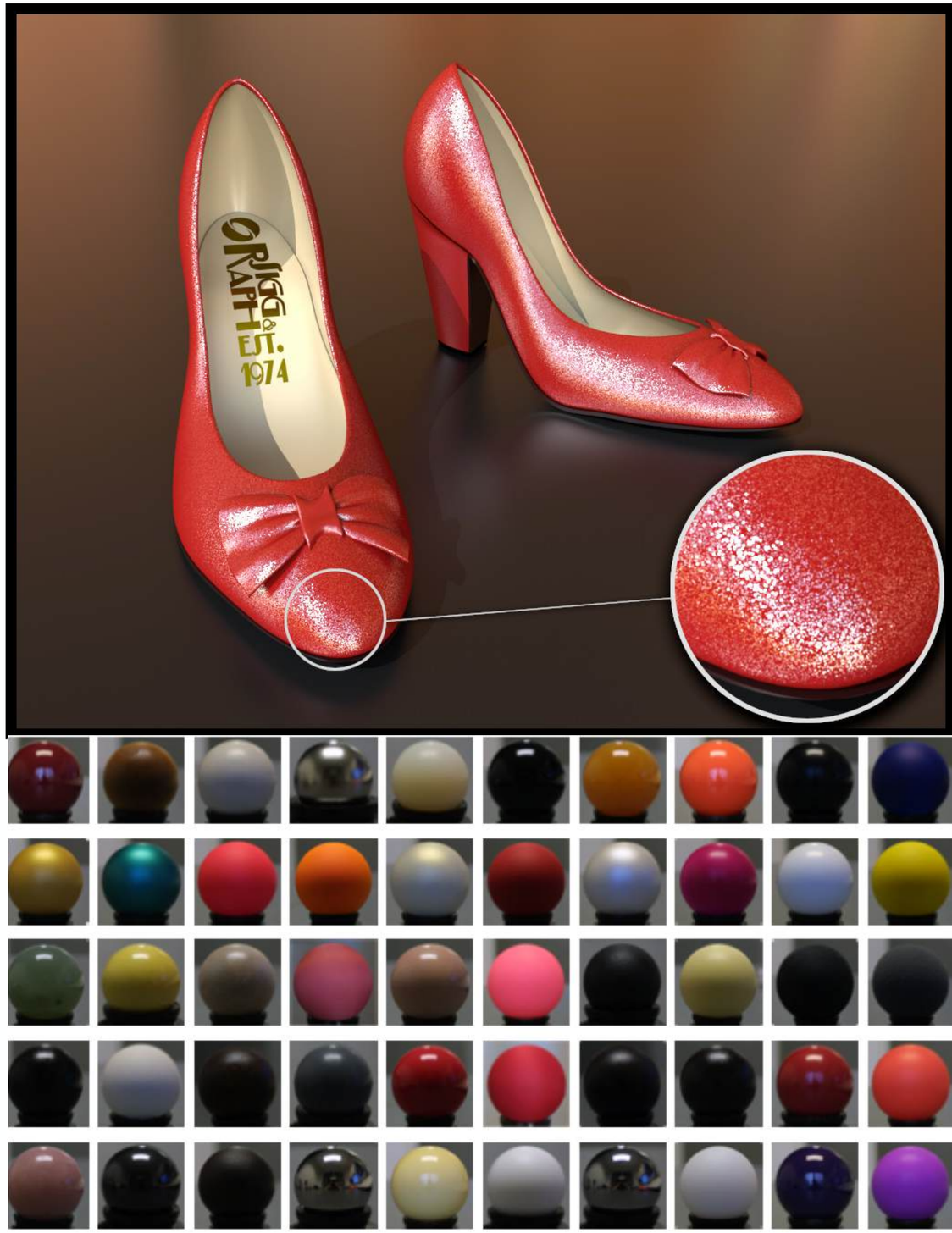






Modeling Material Properties





Modeling Lighting



Realistic Lighting Simulation



Light Transport and Image Synthesis





Photograph (CCD) vs. computer rendering

How Do Cameras Work?



Glenn Derene, Popular Mechanics

Motion Capture

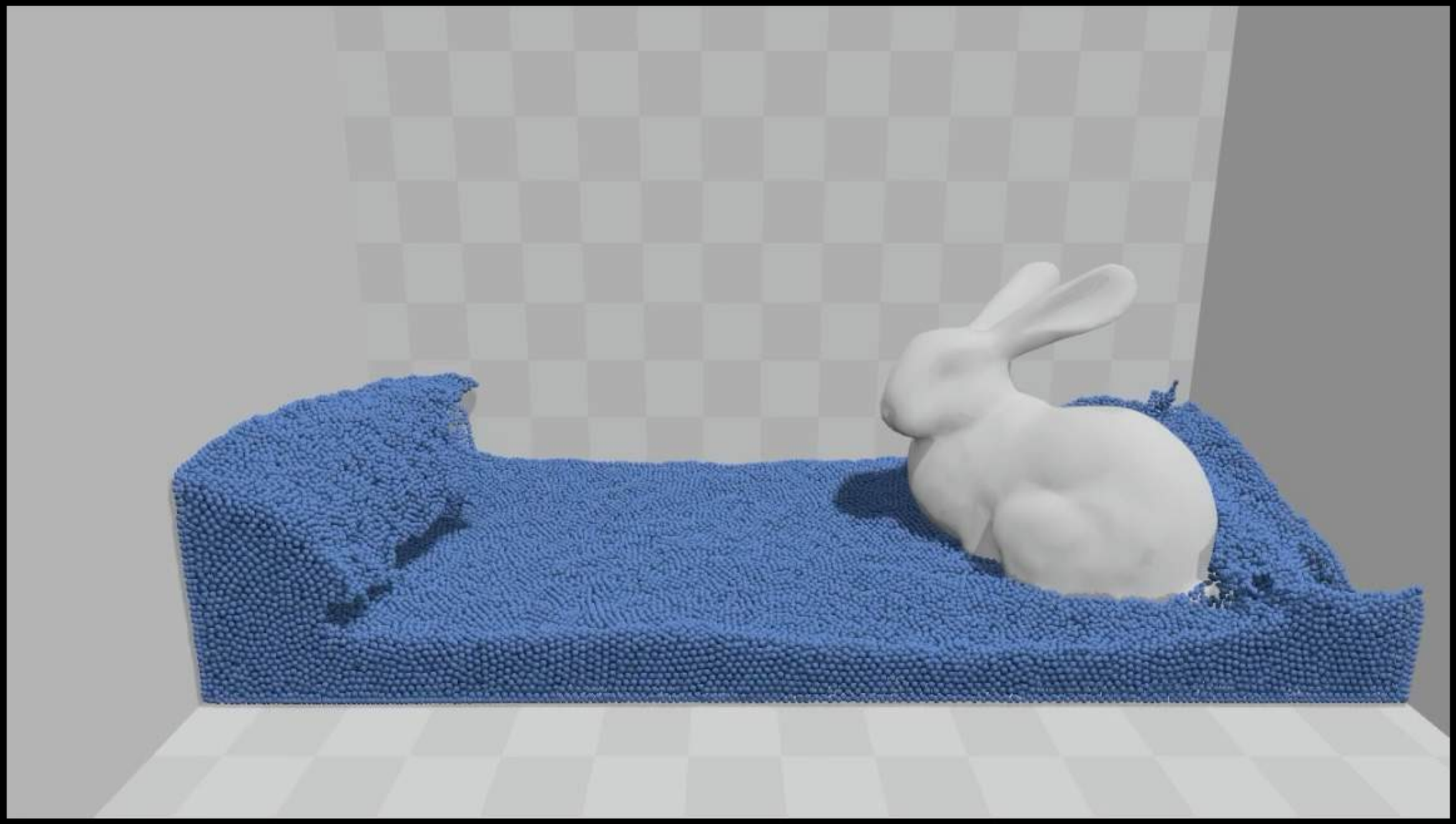


Andy Serkis in The Two Towers



Avatar (2009)

Animation and Physical Simulation



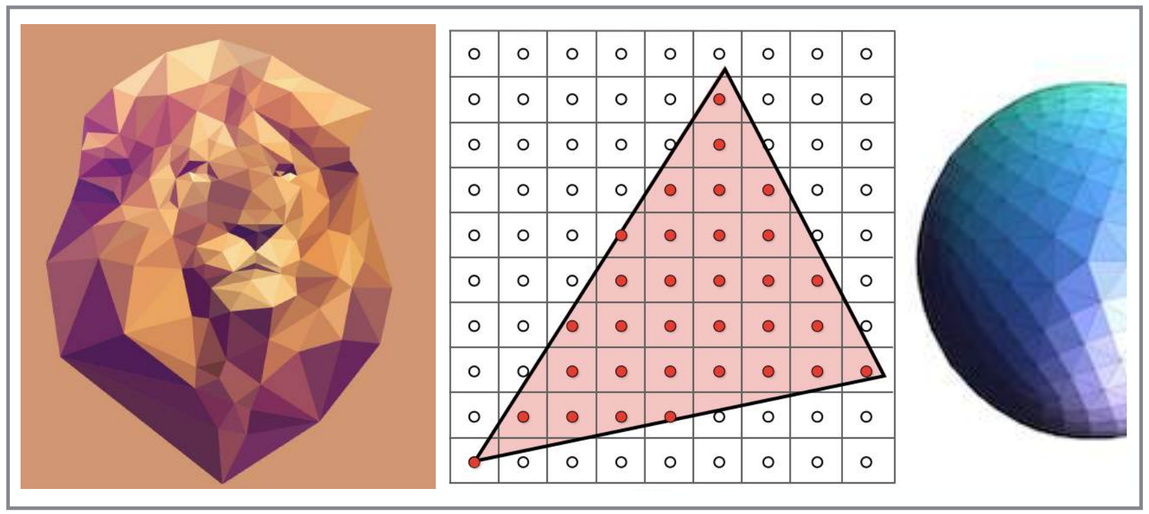
Position Based Fluids, Macklin and Müller

Virtual Reality

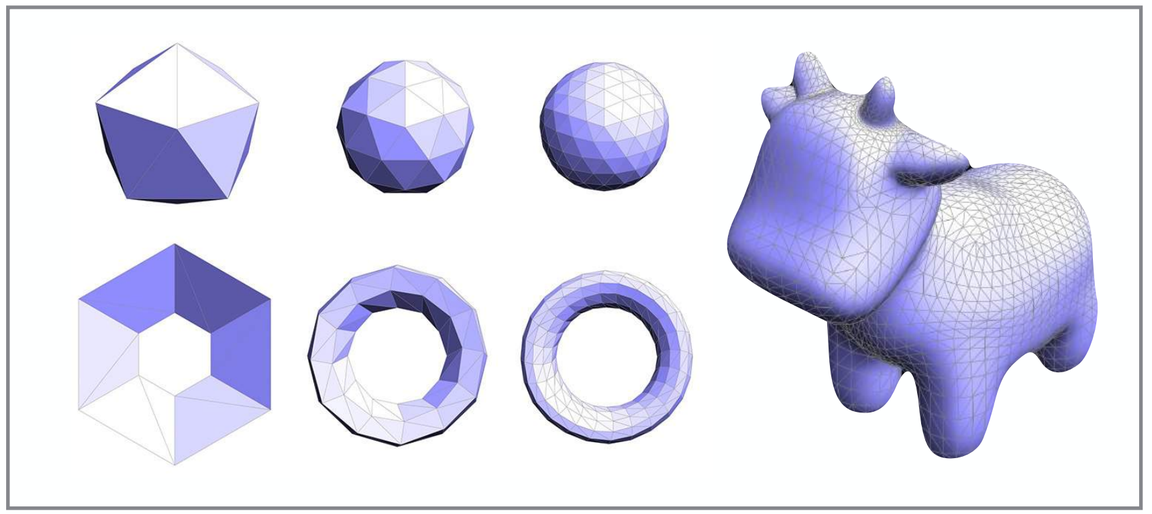


Hands-On Learning

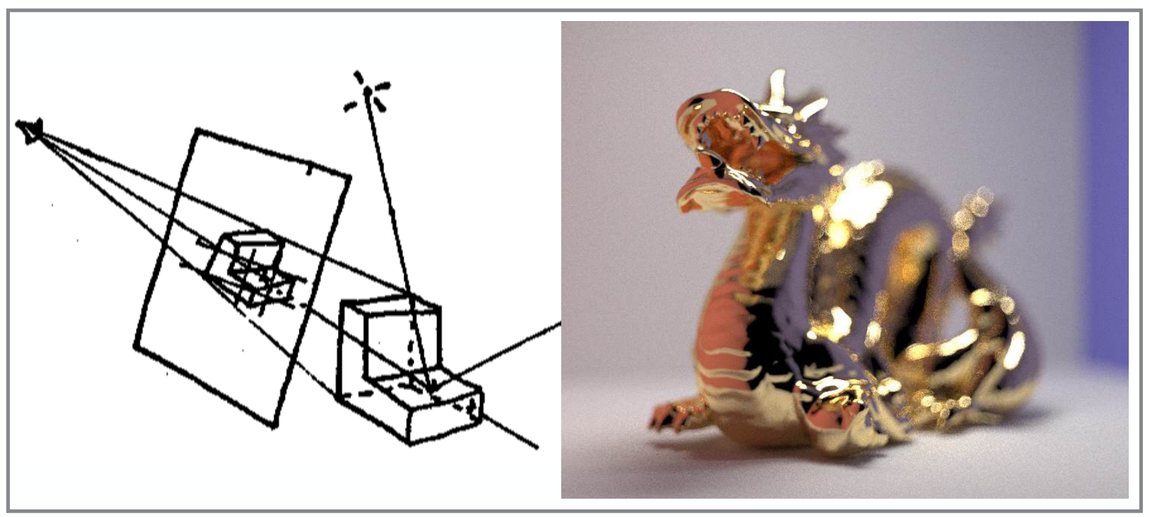
Course Assignments



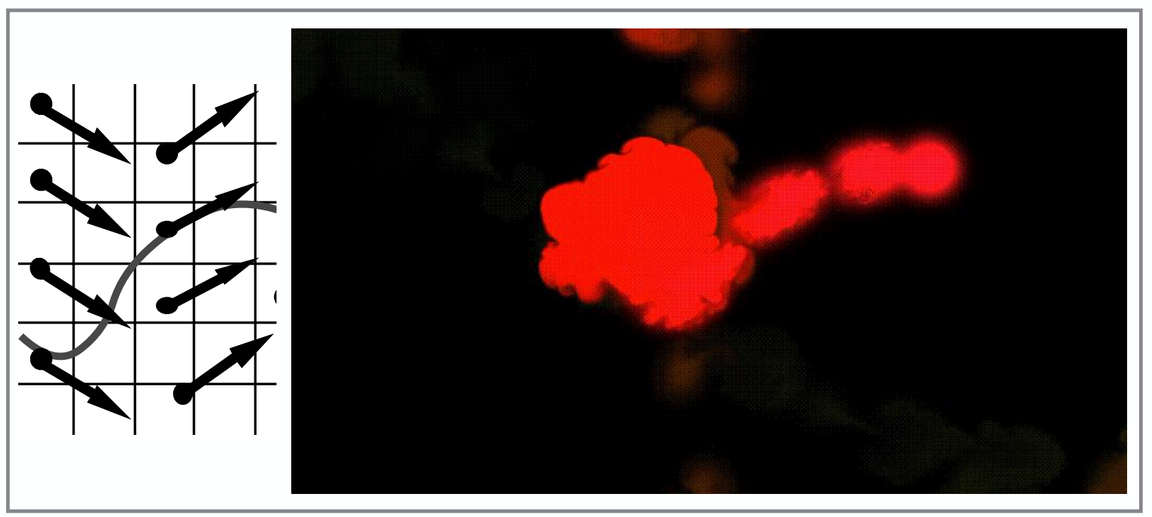
1. Digital Drawing (2 weeks)



2. Geometry (2 weeks)



3. Ray-Tracing (4 weeks)



4. Animation (2 weeks)

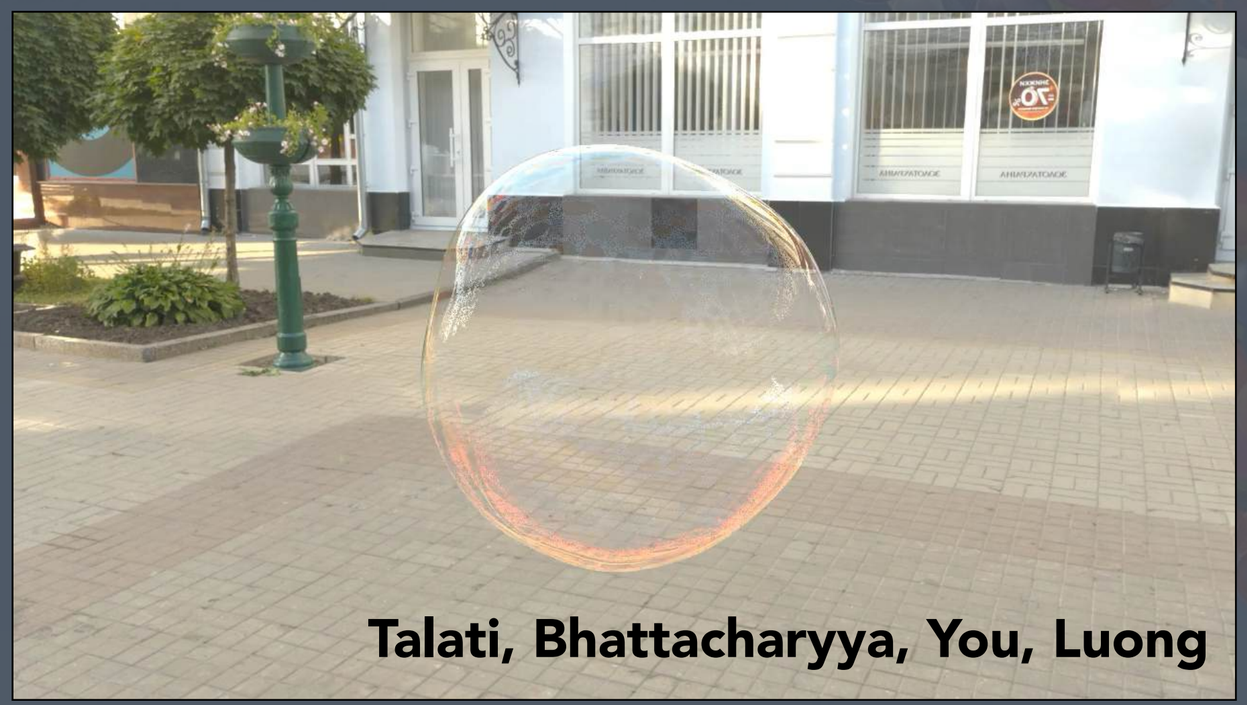
Final Project

Project Competition

- 4 weeks, let your creativity take flight!
  (we will have suggested projects)
- Proposal; checkpoint; presentation, video, report

Credit: Pixar, Up

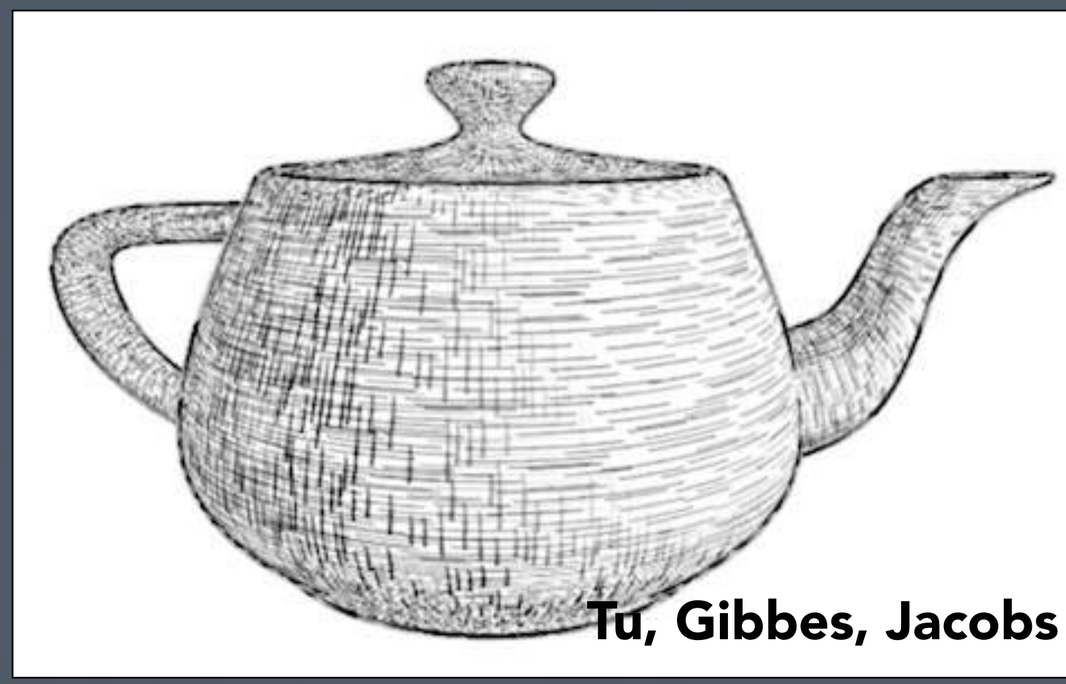
Final Project - Examples



Talati, Bhattacharyya, You, Luong



Ni, Wu, Yu, Zhou



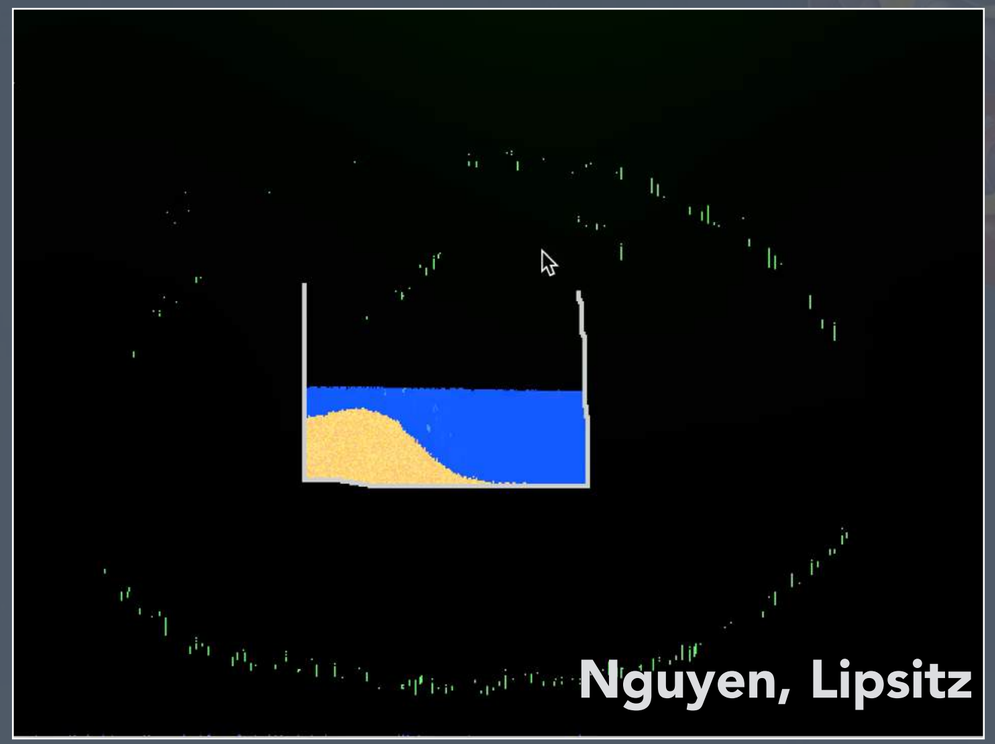
Tu, Gibbes, Jacobs

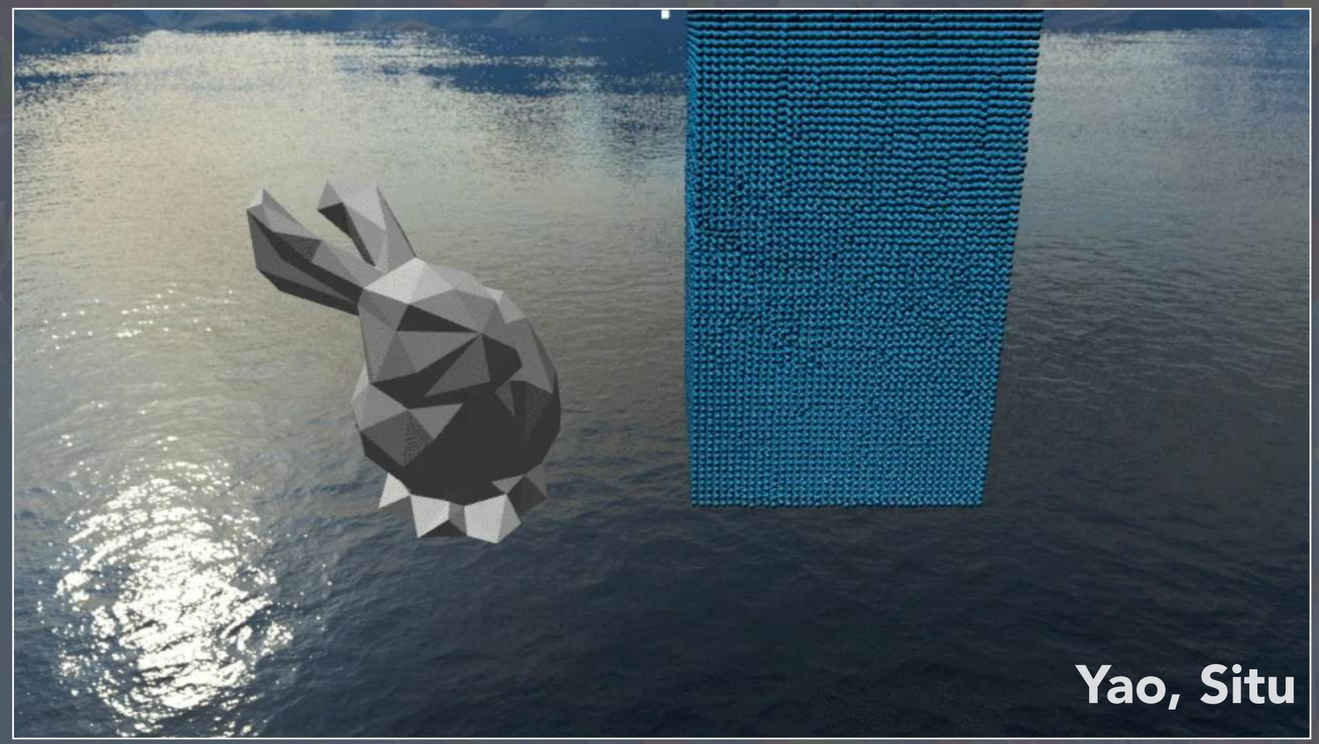


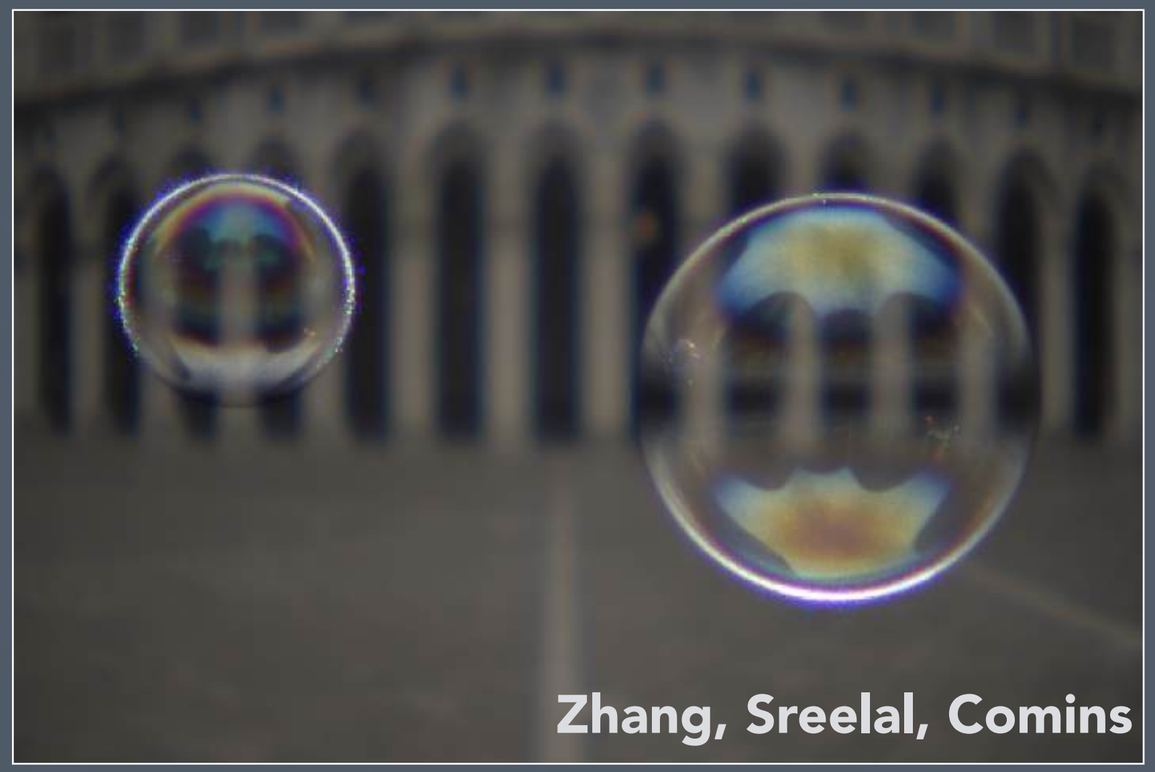
Doriwala, Kamat, Lim, Feguson

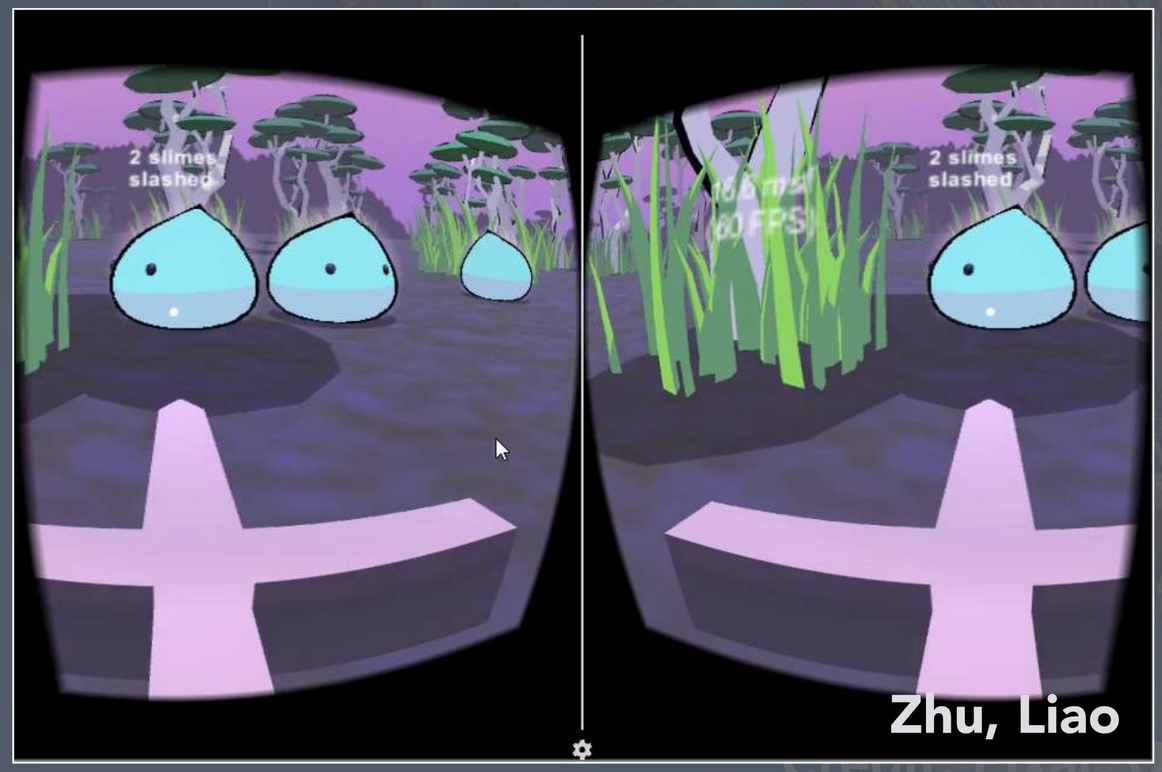
More examples and project reports on prior year course websites: cs184.eecs.berkeley.edu

Final Project - Examples



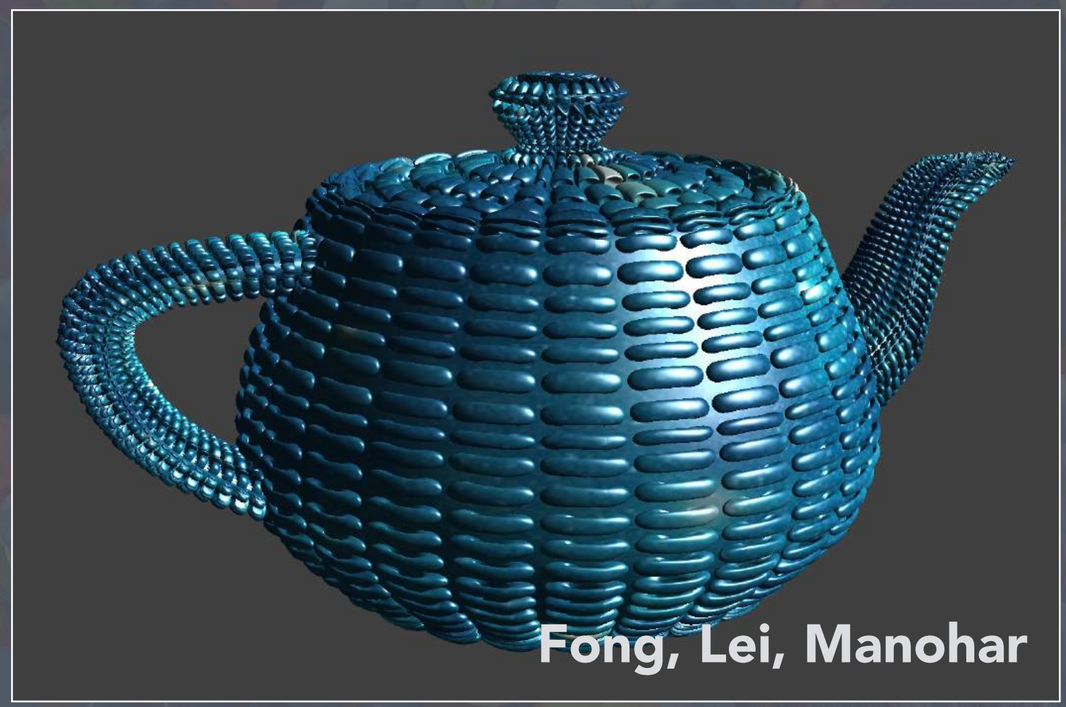


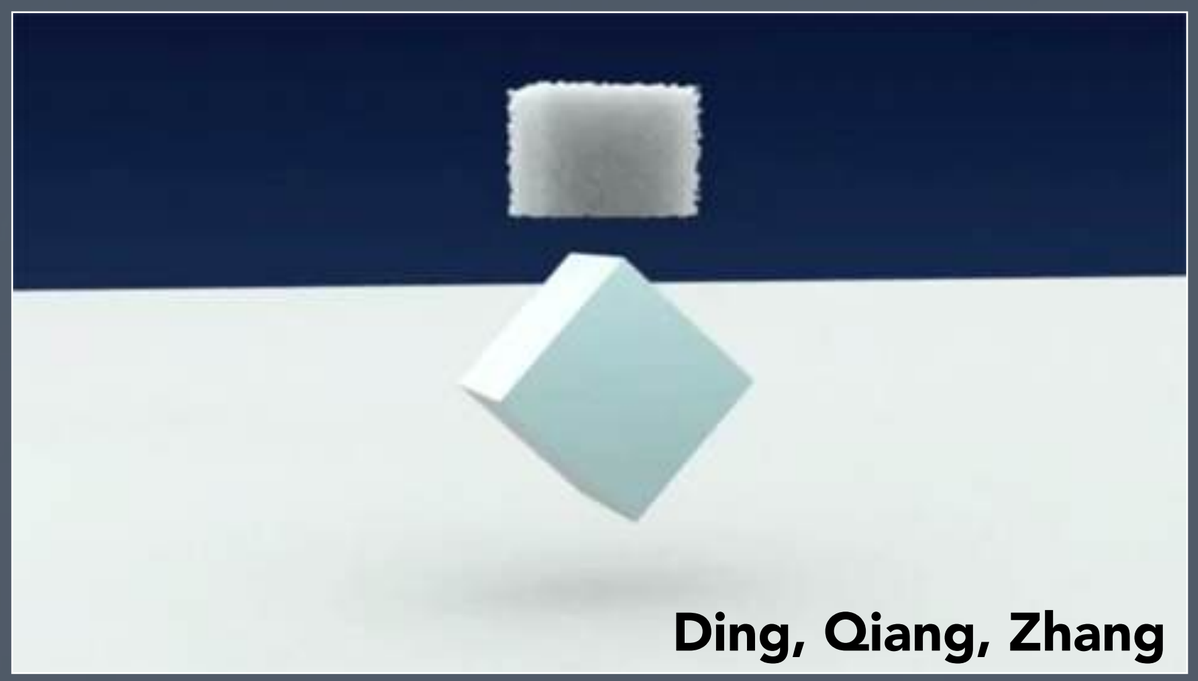


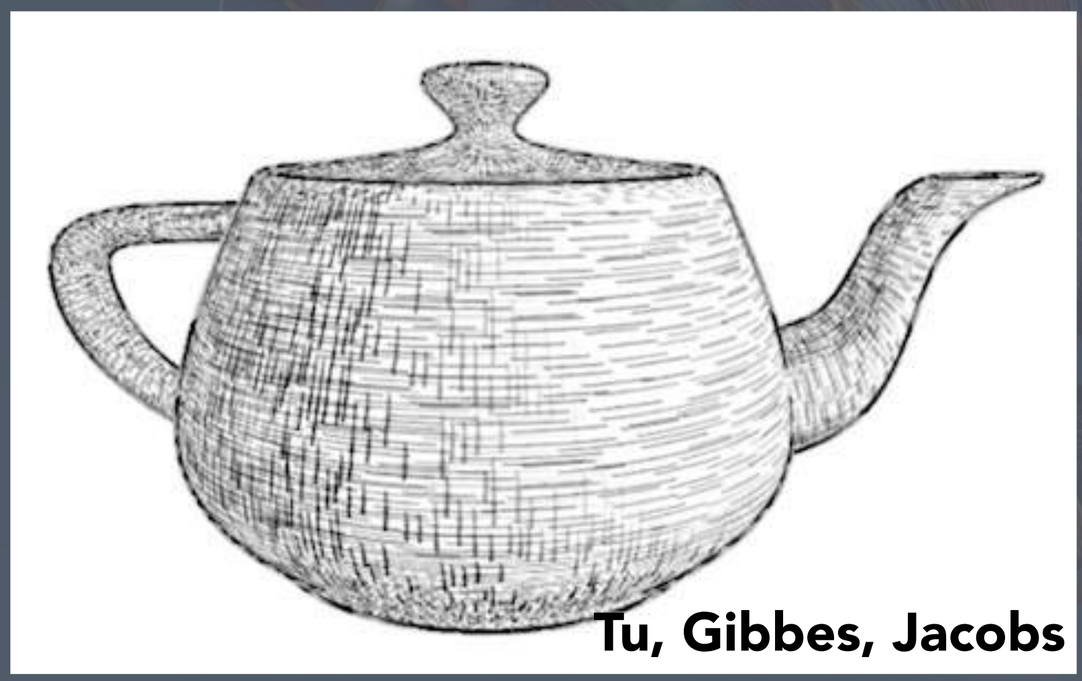


Final Project - Examples



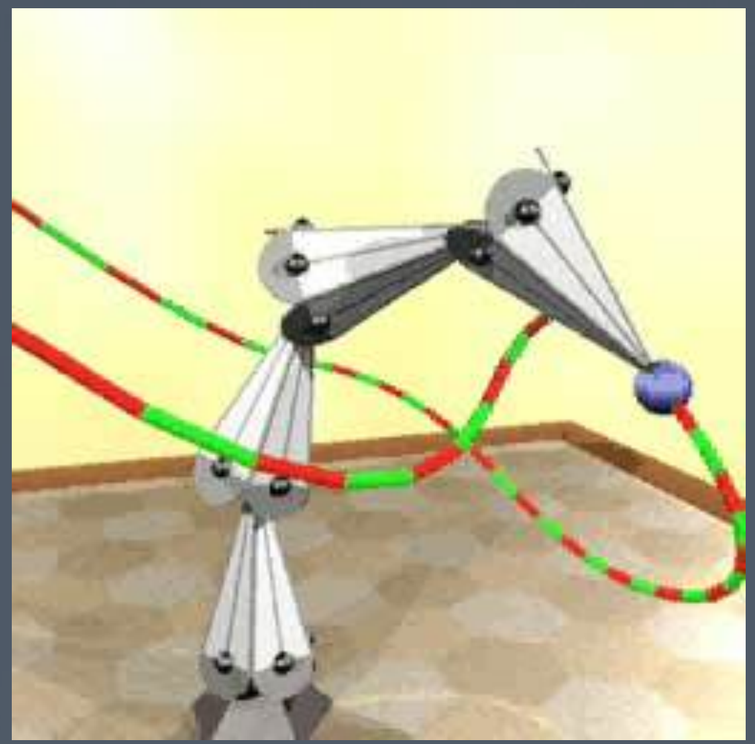






Credit: Pixar, Up

Final Project - Examples









Egon Pasztor

Credit: Pixar, Up

Course Logistics

Zoom

Classes and Sections will be online

- Zoom info is in B-Courses
- Requires that you're logged in with UC Berkeley credentials

Enrollment

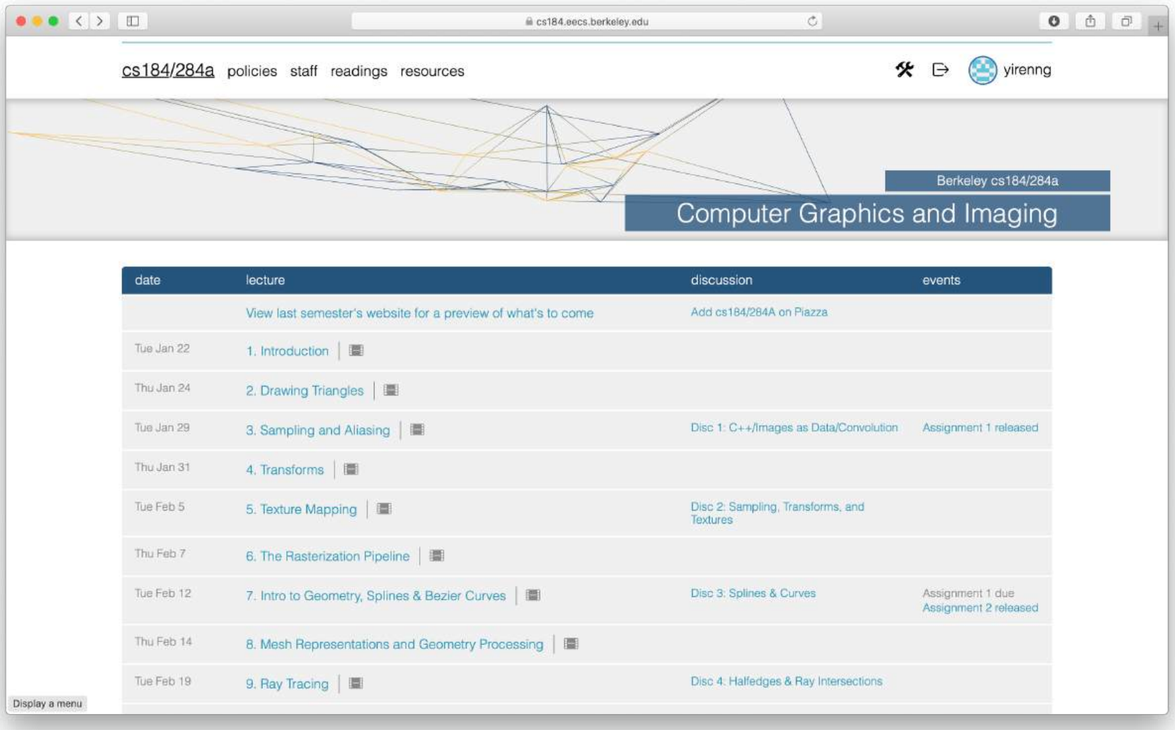
- Past years, high turnover from wait list
- Questions about enrollment:
  - CS184: ask scheduler, cs-scheduling@berkeley.edu
  - CS284A: contact instructors on Ed

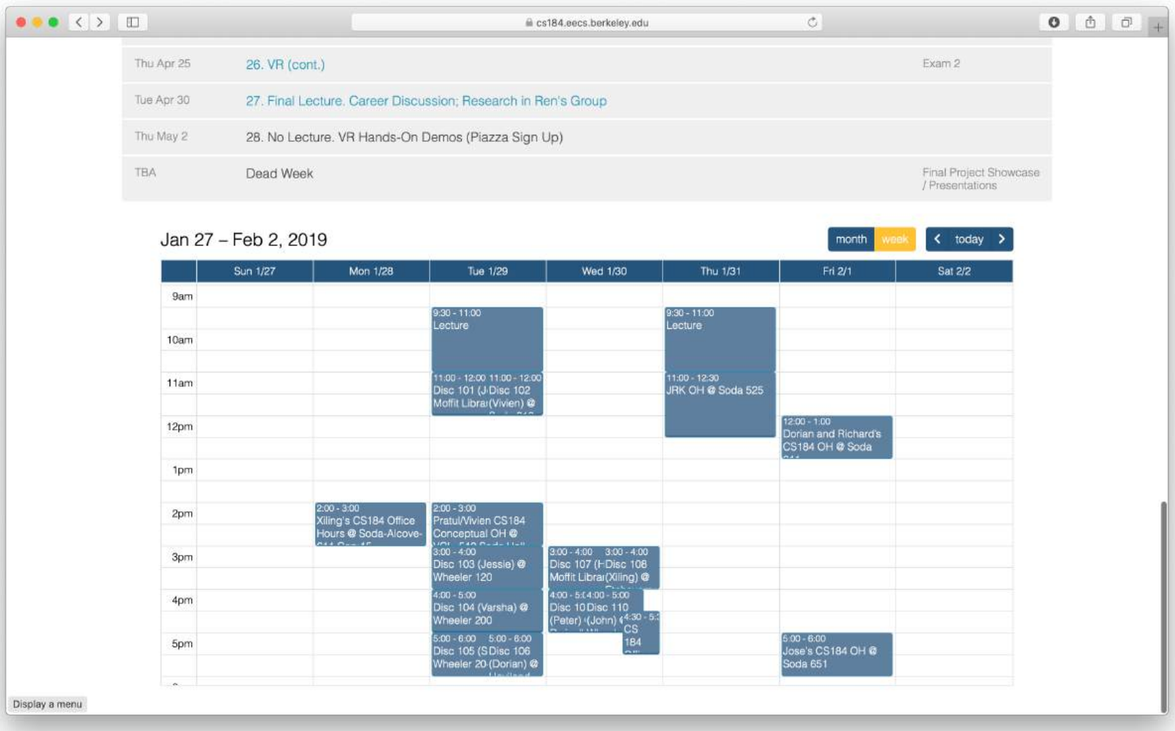
Course Schedule

cs184.eecs.berkeley.edu

Full schedule for class will be on website soon

Note class calendar at bottom for office hours, homework, parties, etc.

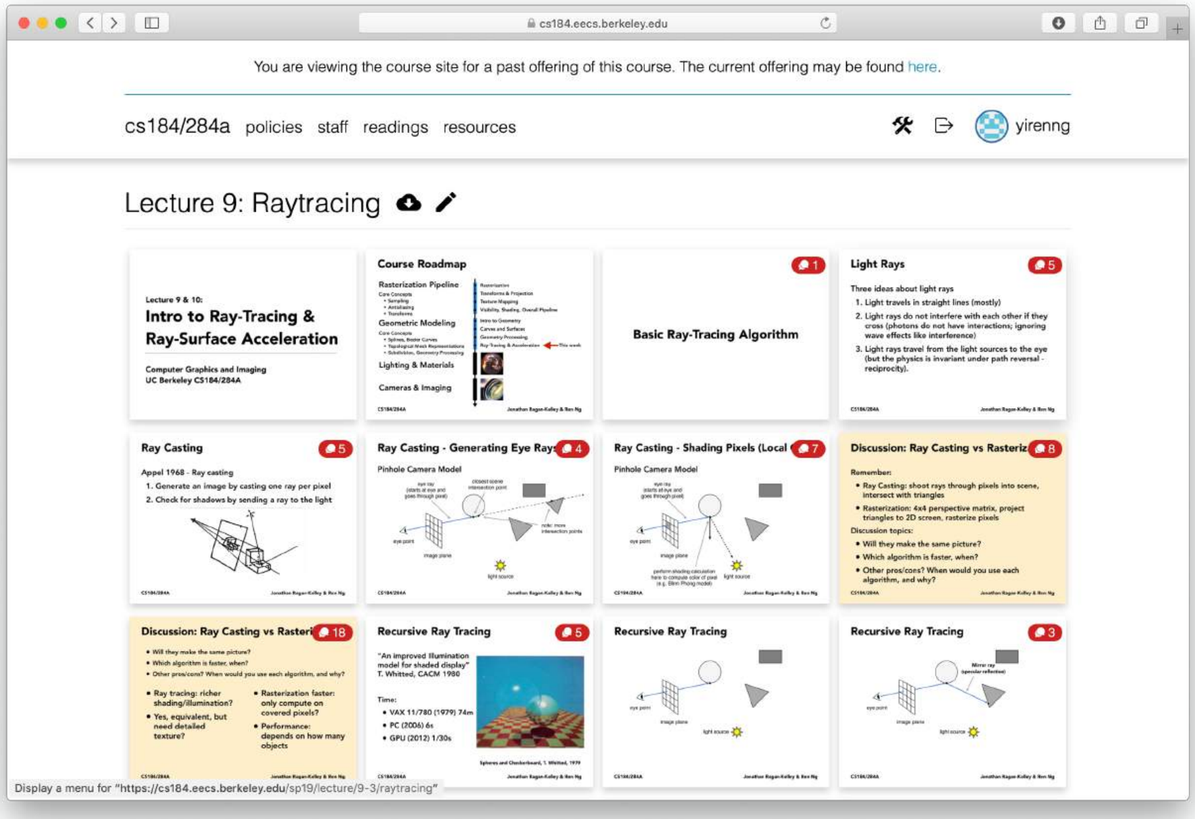




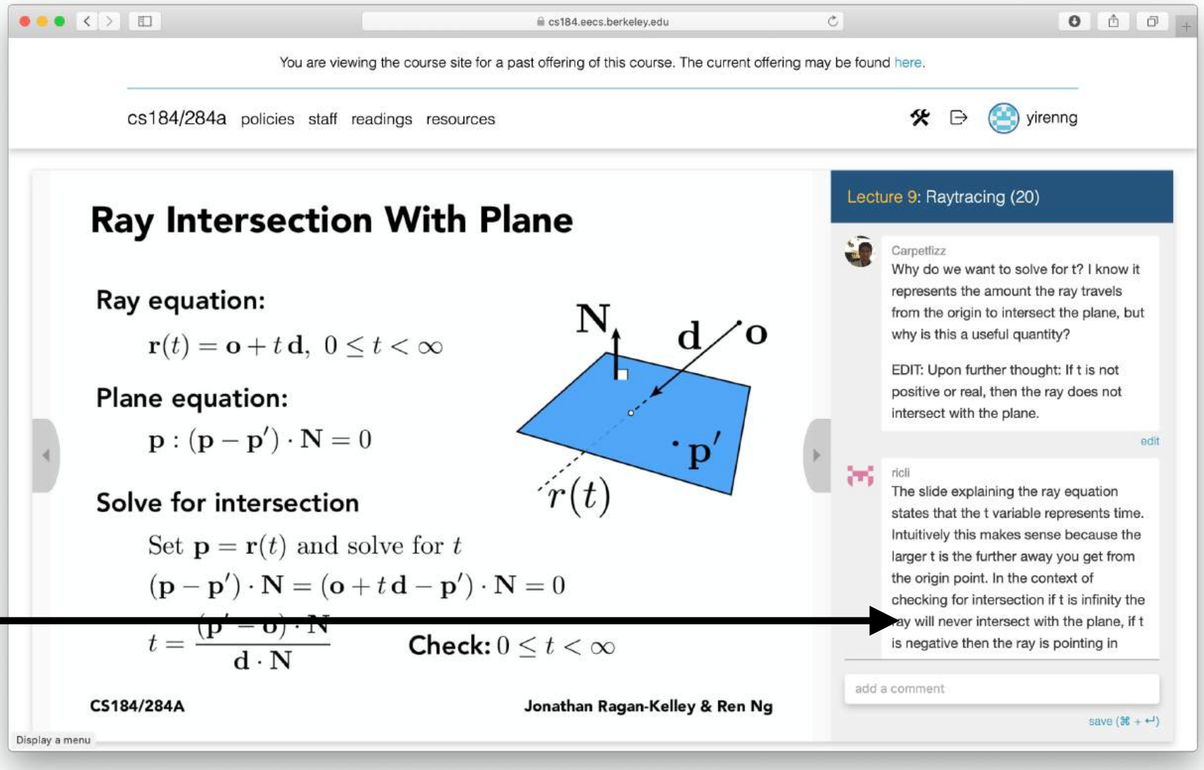
Lecture Slides

cs184.eecs.berkeley.edu

Lecture slides and instructor/TA/ student discussions on the web are the primary course reference materials



Slide comments and discussion



CS184/284A

S26 - O'Brien

Ed

edstem.org

You should be added already (if not, please sign up)!

For logistics and general communication / discussion

- Please use Ed instead of email
- But intellectual discussions about content should primarily go on website as slide comments

Section

Sections start next week, and TAs will give a primer on C++ and building class projects

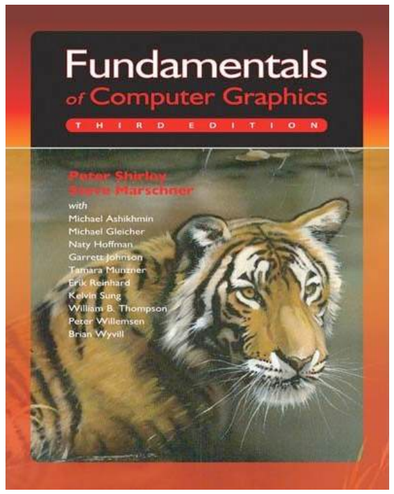
Resources

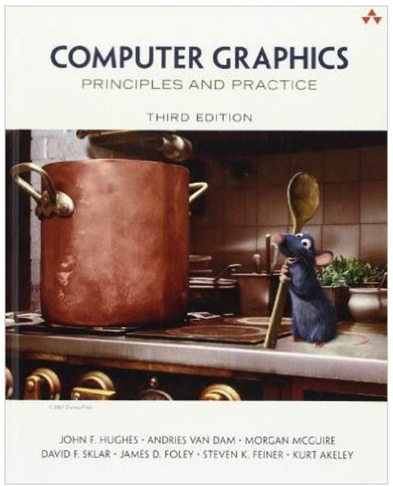
Lectures will be primary source

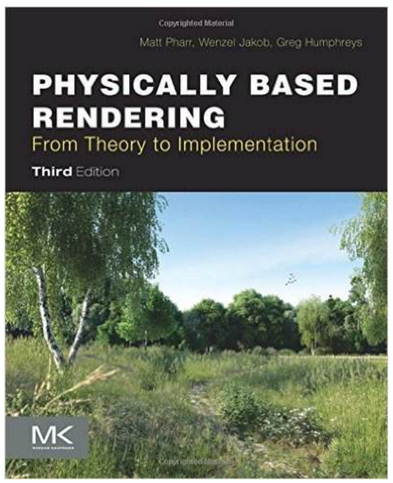
Textbook reference material (optional):

- Fundamentals of Computer Graphics
  by P. Shirley, S. Marschner, et al.
- Computer Graphics: Principles and Practice
  (3rd Edition)
  by Hughes, van Dam, et al.
- Physically Based Rendering, Third Edition:
  From Theory to Implementation
  by Pharr, Jakob and Humphreys

Other optional reading resources on class website







Learning, Grading, Collaboration & Culture

Goals:

- Enable you to increase focus on learning rather than assessment
- Encourage your learning through collaboration
- Entrust you with maintaining academic integrity

Main Ideas (details on course website — please review):

- The class is not graded on a curve.
- Collaboration in pairs encouraged on programming assignments.
- Exams will be take-home, with honor code, no proctoring.

Details

- Please read the Policies page on the course website and ask questions on Ed.

Course Deliverables and Assessment

CS184: your course grade is out of 100 total points

- Five homework assignments, 10 points each
  - Pair projects encouraged. Programming and written reports.
- Two exams, 10 points each
  - Check dates on website schedule. No exam during Finals Week.
- Final project, 25 points
  - In groups of four, with final presentation, video, report.
- Participation, 5 points
  - Attend lectures/discussion, and/or write website comments on lecture slides.

CS284A students: Project is 40% of grade, remainder normalized.

What We Are Looking For In Slide Comments

Try to explain the slide (as if teaching your classmate to study for an exam)

- "Ren said this, but if you think about it this way it is much clearer"

Explain what is confusing you

- "What I was totally confused about here was…"

Challenge classmates with a question

- For example, make up a question you think might be on an exam

Provide a link to an alternative explanation

- "This site has a really good animation of pre-filtering to avoid aliasing"

Mention real-world examples

- For example, describe what default interpolation functions are used in iOS.

Constructively respond to another student's question

- "@nojaggies, are you sure that is correct? I thought that Ren said…"

It is OK, and even encouraged, to address the same topic (or repeat someone else's summary, explanation or idea) in your own words

- "@cornellbox's point is that subdivision is also used to…"

Late Days Policy

Assignments are late after 11:59pm on due date.

You have 3 late days per assignment (not final project)

- Extend a homework assignment deadline by 24 hours using one late day.
- If you do not have remaining late days, 1 point penalty per day.
- Please use this flexibility to manage your exceptional circumstances.

Class Philosophy

We want to build an active, engaged class community.

Come to class, participate in lecture, discussion, office hours, homework parties.

Practice cooperative, supportive learning.

Contribute on the website.

Uphold academic honor individually and collectively.

Inclusive Classroom

Respect each other as individuals with unique identities and backgrounds.

Help create a welcoming community for our full diversity of perspectives and experiences.

Collaboration and team learning are encouraged, and will be supported through class staff and resources.

Projects are a great way to meet new people and make friends; work on building trust and leveraging each other's unique strengths.

Course Roadmap

Rasterization Pipeline

Core Concepts

- Sampling
- Antialiasing
- Transforms

Geometric Modeling

Core Concepts

- Splines, Bezier Curves
- Topological Mesh Representations
- Subdivision, Geometry Processing

Lighting & Materials

Core Concepts

- Measuring Light
- Unbiased Integral Estimation
- Light Transport & Materials

Cameras & Imaging

- Rasterization
- Transforms & Projection
- Texture Mapping
- Visibility, Shading, Overall Pipeline
- Intro to Geometry
- Curves and Surfaces
- Geometry Processing
- Ray-Tracing & Acceleration
- Radiometry & Photometry
- Monte Carlo Integration
- Global Illumination & Path Tracing
- Material Modeling



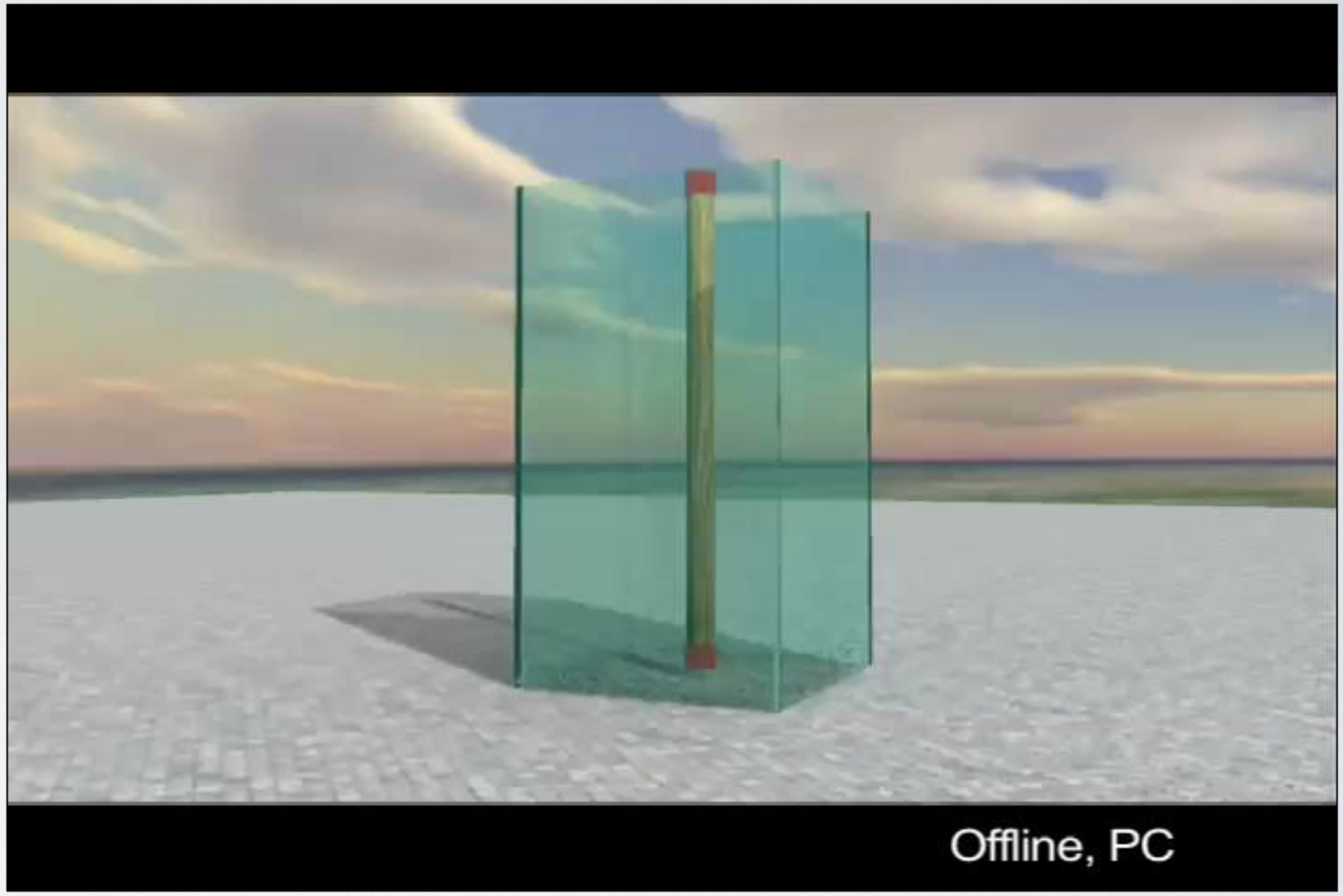
My research

- Simulation

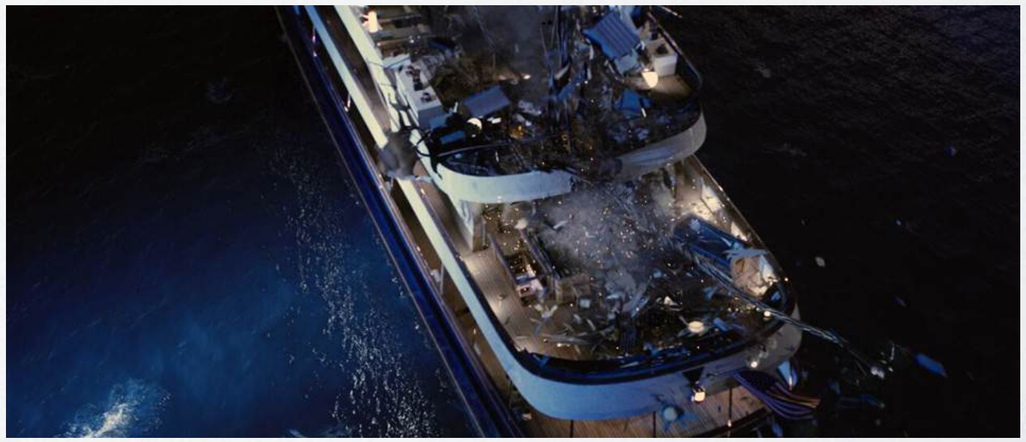








With Eric Parker



My research

- Simulation



With Huamin Wang
and Ravi Ramamoorthi

My research

- Simulation

With Rahul Narain and
Armin Samii

Adaptive Anisotropic Remeshing for Cloth Simulation

Rahul Narain, Armin Samii, James F. O'Brien
University of California, Berkeley

SIGGRAPH Asia 2012

My research

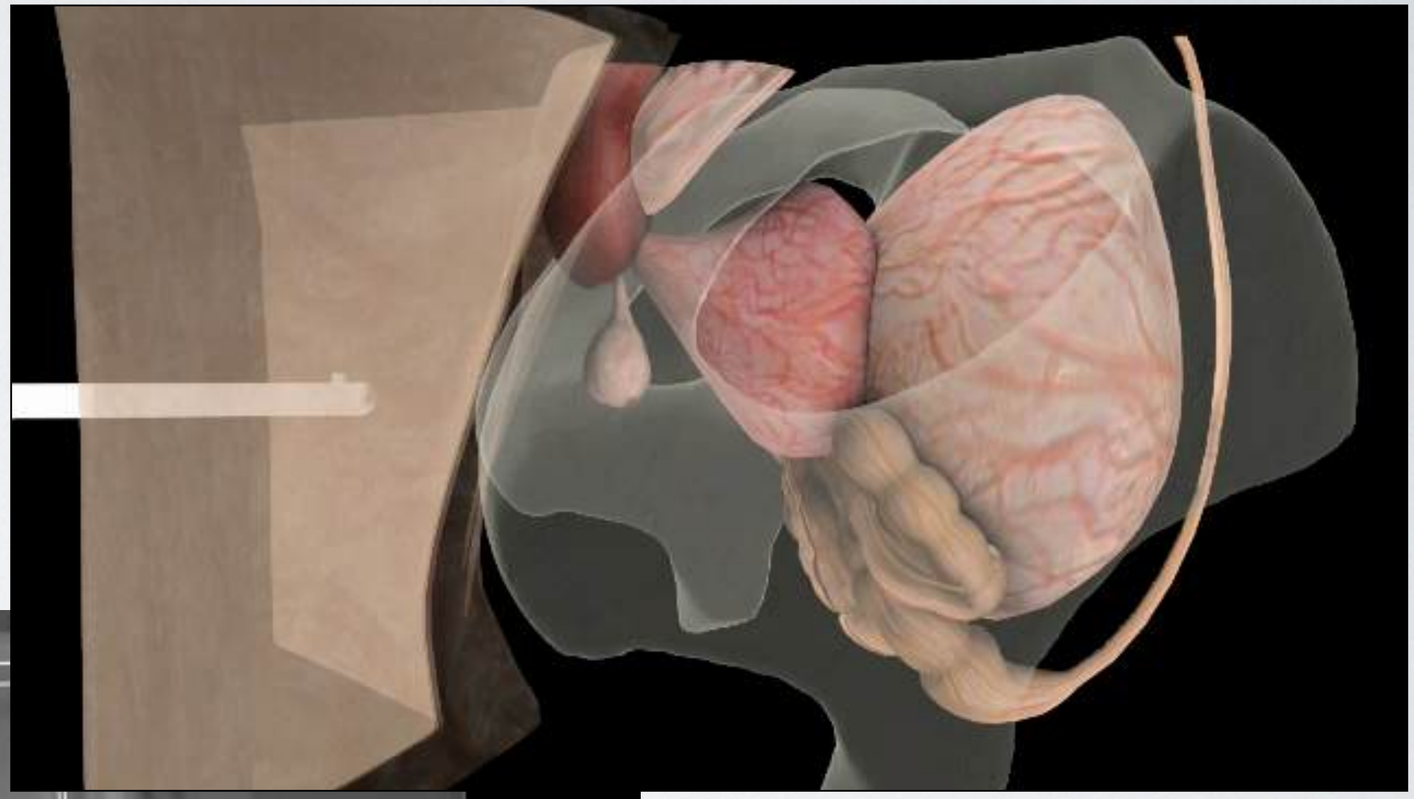
- Simulation

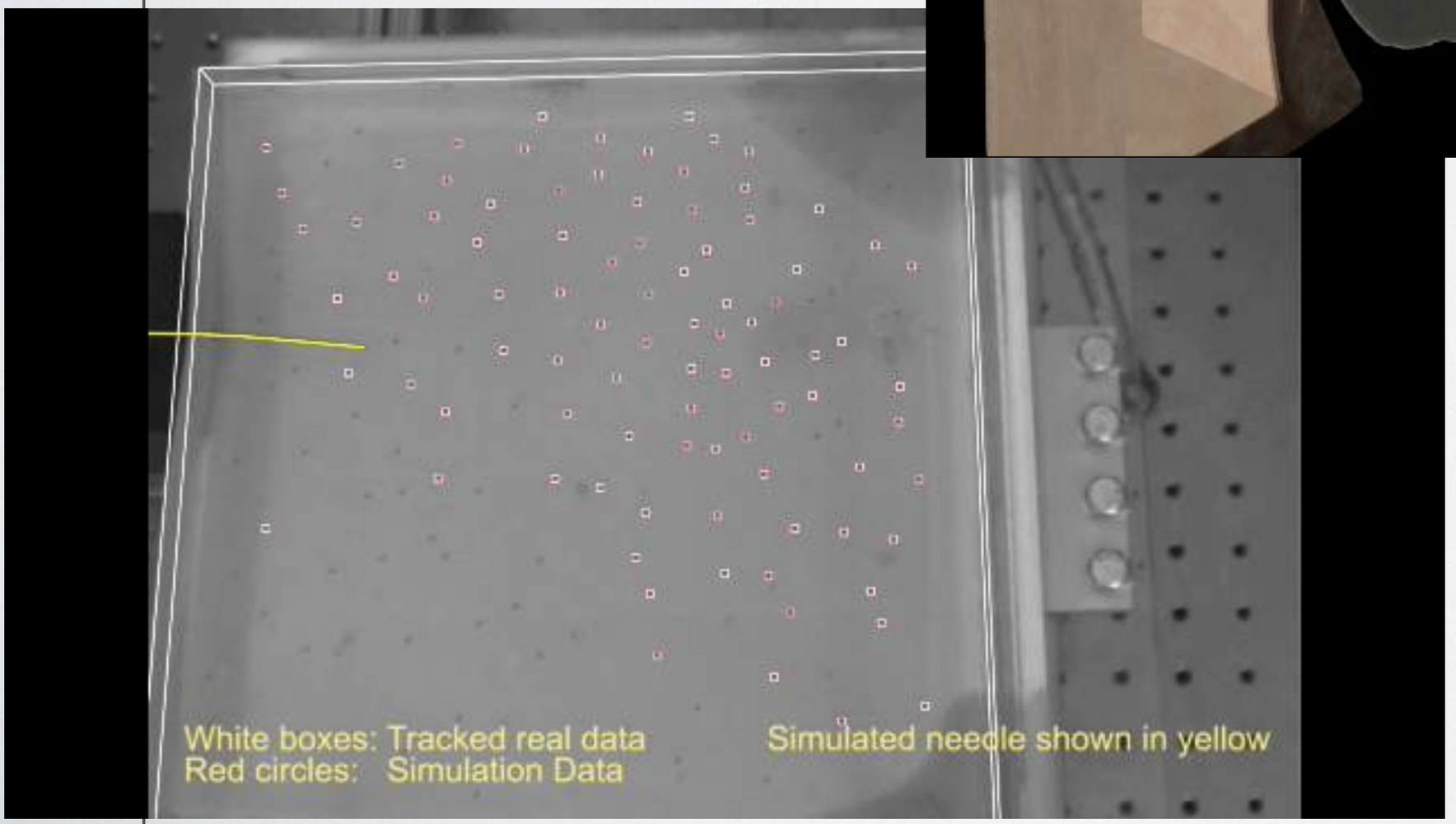
With Rahul Narain and
Tobias Pfaff



My research

- Simulation





With Nuttapong Chentanez, Ron Alterovitz, Daniel Ritchie, Lita Cho, Kris K. Hauser, Ken Goldberg, and Jonathan R. Shewchuk

My research

- Simulation

With Adam Kirk



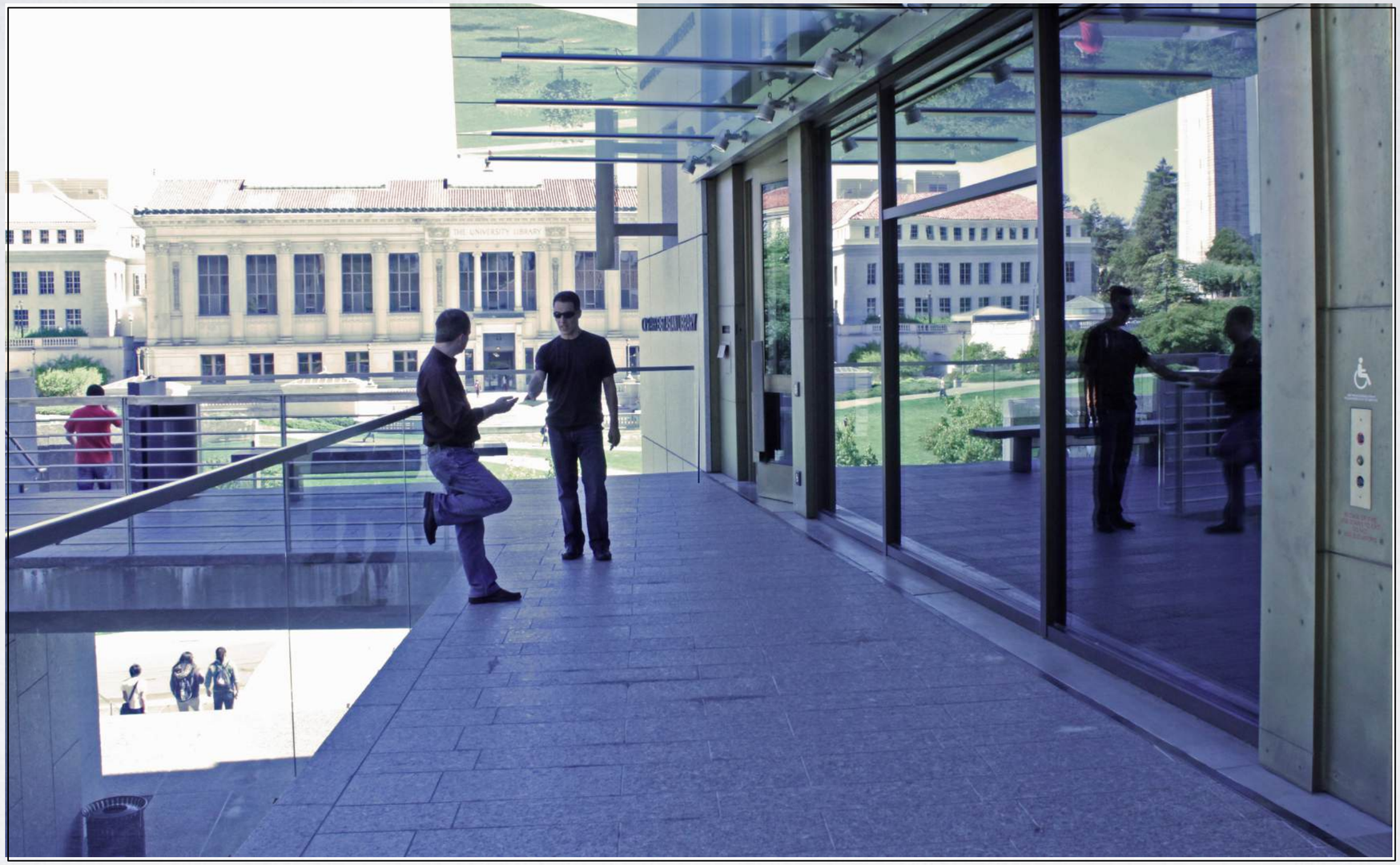




My research

- Forensics

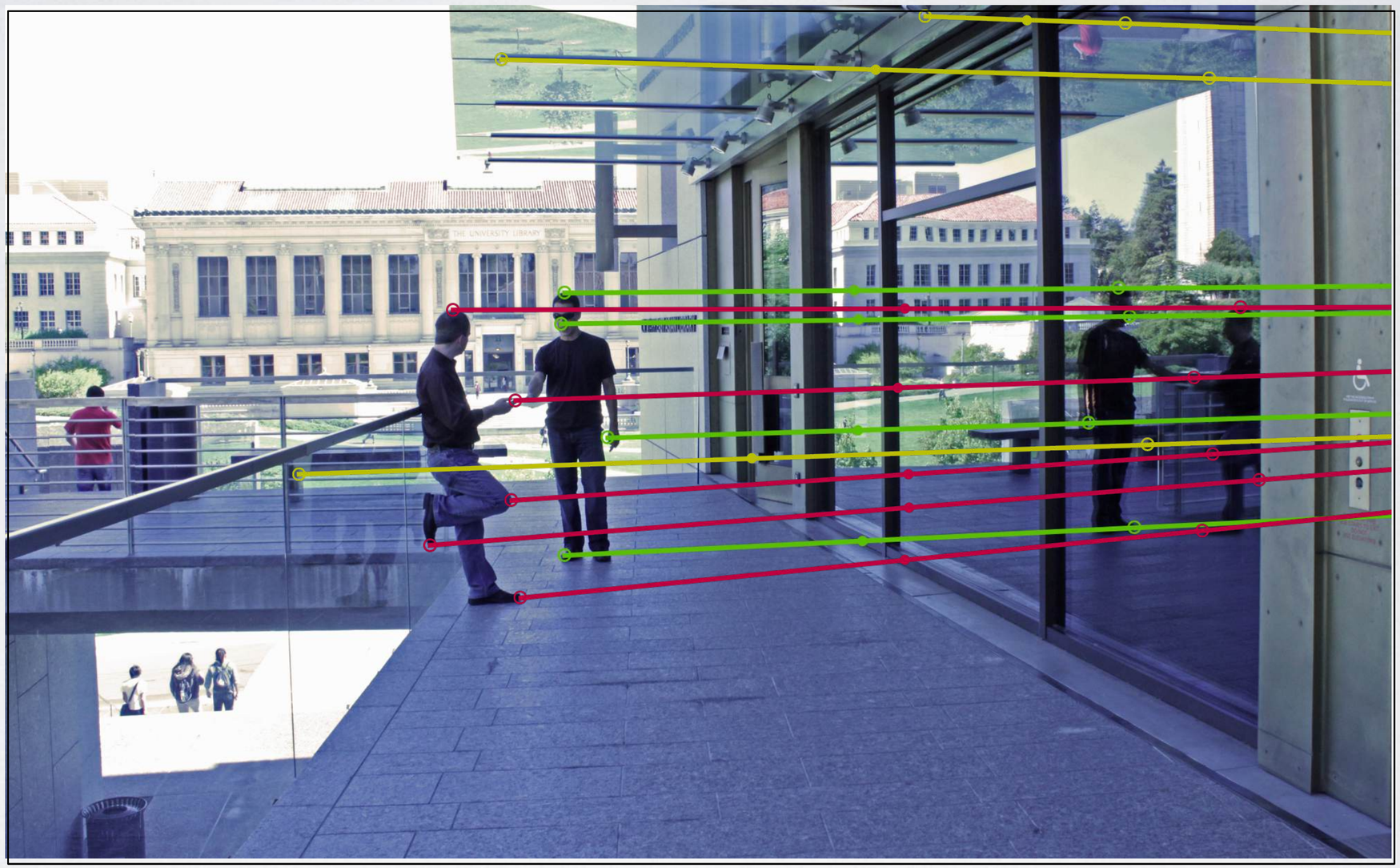
With Hany Farid

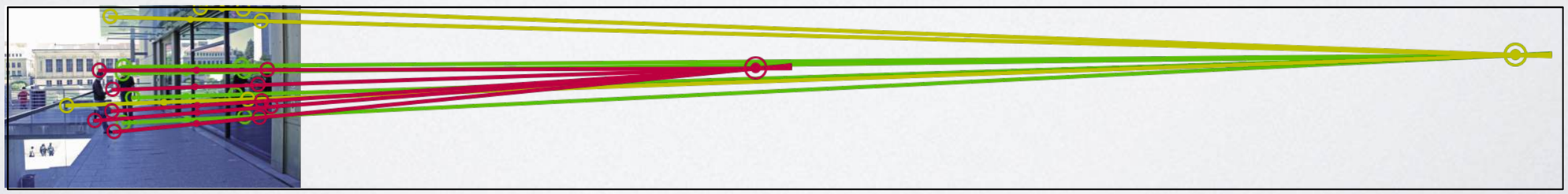


My research

- Forensics

With Hany Farid





My research

Adaptive Tearing and Cracking of Thin Sheets
Tobias Pfaff, Rahul Narain, Juan Miguel de Joya, and James F. O'Brien

With Tobias Pfaff, Rahul Narain,
and Juan Miguel de Joya,



SIGGRAPH 2020

Fast and Deep Facial Deformations

Stephen W.. Bailey, Dalton Omens, Paul DiLorenzo and James F. O'Brien

Fast and Deep Facial Deformations

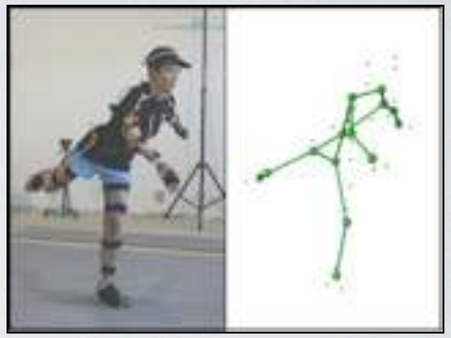
Stephen W. Bailey UC Berkeley and Unity Technologies
Dalton Omens UC Berkeley
Paul DiLorenzo DreamWorks Animation
James F. O'Brien UC Berkeley

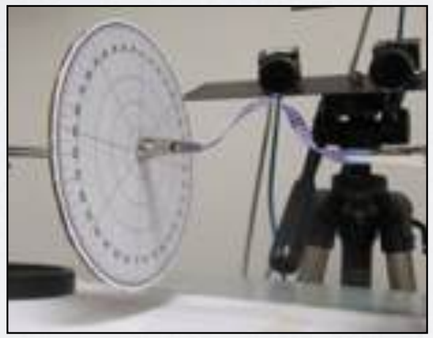
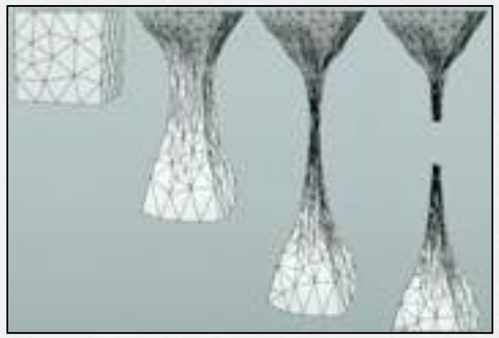
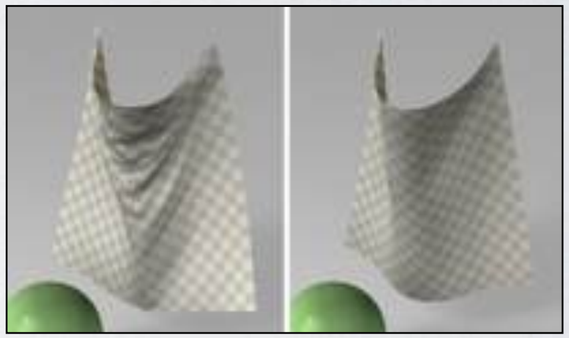


http://obrien.berkeley.edu

See my webpage for more information and other projects...












Acknowledgments

Thanks to Ren Ng, Angjoo Kanazawa, Pat Hanrahan, Kayvon Fatahalian, Keenan Crane, and Mark Pauly for presentation resources.